Allocating Influence: Government Advertising in Mexico's News Market

Camila Blanes*

November 9, 2026

Abstract

Governments, whether autocratic or democratic, seek to influence the media, typically through censorship, bribes, or quid pro quo arrangements. I provide evidence of a distinct channel of influence that operates through market competition, namely political favoritism in the allocation of public resources. By directing public resources toward ideologically aligned outlets, a government alters the relative strength of ideologically opposed outlets in the market for news. Combining the universe of federal advertising contracts in Mexico with the full text of national print media, I exploit the 2018 transition from a right-wing to a left-wing administration and show that aligned outlets receive disproportionately more advertising than misaligned outlets. These allocations reflect favoritism rather than the audiences a government would need to reach, such as the incumbent's own voters. I then document the consequences in the media market, where aligned outlets differentially increase their news production and original reporting relative to misaligned outlets. Media influence can therefore operate through competitive advantages in the media market rather than through censorship or quid pro quo.

Keywords: Government advertising; Media capture; Political favoritism; Ideological alignment
JEL Codes: D72, L82, M37, D73

*Stanford Graduate School of Business, Political Economy. Email: cblanes@stanford.edu. I am grateful to Greg Martin, Matthew Gentzkow, and Horacio Larreguy for their guidance. I also thank Beatriz Magaloni, Saumitra Jha, Ali Yurukoglu, Stephanie Zonszein, Jose Ramon Enriquez, Sam Asher, Matthew Merrigan, and Janet Malzahn for helpful comments and suggestions, as well as participants at APSA (2026), the XXIII RIDGE/LACEA Workshop on Political Economy, and the 9th Economics of Media Bias Workshop.

1 Introduction

An independent press is central to the functioning of the democratic process because it informs voters about the performance of their representatives and enables them to hold the government accountable (Besley and Prat, 2006; Snyder Jr. and Strömberg, 2010). Government control of the media is among the most common means of undermining democracy. Governments can exert such control through censorship, propaganda, bribery, and quid pro quo arrangements, practices documented primarily in autocratic regimes, competitive authoritarian regimes, and backsliding democracies (King et al., 2013; Adena et al., 2015; Enikolopov et al., 2011; McMillan and Zoido, 2004; Kronick and Marshall, 2024; Szeidl and Szucs, 2021).

However, evidence from consolidated democracies remains limited, partly because two forces constrain government influence in these settings. Constitutional guarantees protect the freedom of the press and raise the legal and political cost of direct interference with news content. Market discipline enables readers to sanction outlets that provide biased coverage by switching to competing outlets, so an outlet that distorts its coverage loses part of its audience and the advertising revenue that audience generates (Besley and Prat, 2006; Durante and Knight, 2012). A government that seeks favorable coverage must therefore compensate the outlet for these losses, and the required compensation rises with the outlet's commercial revenue (Gehlbach and Sonin, 2014). Governments can nonetheless influence the media through indirect channels.

This paper documents an alternative mechanism of media influence that operates through political favoritism in the allocation of government advertising. Such advertising is a major source of revenue for media outlets in many developing and developed countries, particularly where private advertising markets are either limited or concentrated.[1] Governments can use the selective distribution of these resources to alter the relative

[1]Countries where the government is a major advertiser in the market include Bulgaria, Greece, Hungary, Poland, Romania, Slovakia, Slovenia, Albania, Macedonia, Serbia, Turkey, Argentina, Brazil, Colombia, Chile, Mexico, Peru, India, Pakistan, Myanmar.

strength of competing political voices. By channeling funds to ideologically aligned outlets, a government can amplify sympathetic voices, because the additional resources allow those outlets to expand their production of news relative to their competitors.

I provide empirical evidence on the allocation of public resources to the media, and I separate political favoritism from the alternative objectives that could explain the allocation of advertising. I then show how favoritism can become a form of media capture, in which the government gains influence over the media by altering the relative strength of competing political voices. I study this mechanism in Mexico, where the federal government is one of the largest advertisers in a competitive newspaper market. While the stated purpose of government advertising is to promote public programs, highlight achievements, or disseminate prevention messages, its discretionary nature allows incumbents to reward supportive outlets or penalize critical ones. The analysis covers the eighteen national newspapers that account for on average 80.9 percent of national print circulation. For these outlets I assemble a novel dataset combining every federal advertising contract signed between 2014 and 2022, the objective of each campaign and the demographic groups the government states it wants to reach, outlet circulation and advertising prices, and the full text of their daily print editions.

I first show that the government allocates advertising toward ideologically aligned outlets on the extensive and intensive margins. I implement a difference-in-differences specification that leverages the 2018 federal transition from a right-wing to a left-wing administration to compare changes in advertising spending between left- and right-wing outlets. Because advertising relationships persist across administrations, and a ministry's existing outlets may suit its campaigns for reasons unrelated to ideology, I compare the same outlet and ministry over time. I measure outlet ideology from text, using a penalized classifier trained on the editorials of outlets with established ideological positions to place all outlets on a left-right dimension. My point estimates imply that a left-wing outlet becomes 18.3 percentage points more likely to receive a contract from a given ministry,

and receives $105,362 more per ministry-year, than a right-wing outlet after the transition to a left-wing government. The incoming administration cut advertising across these ministries by 48 percent, and that cut fell almost entirely on misaligned outlets, whose spending fell by 72 percent against 15 percent for aligned outlets. Alignment therefore shielded an outlet from a contraction absorbed by its competitors rather than raising what any outlet received.

Next, I show that this reallocation reflects favoritism toward an outlet's ideology rather than the government's efforts to reach different demographic audiences. One alternative explanation is that the reallocation reflects changes in the types of programs being advertised rather than ideology. A left-wing government may promote more social programs whose intended beneficiaries overlap with the readership of left-wing outlets, which would make the reallocation appear ideologically motivated even in the absence of favoritism. A second explanation is that the government prioritizes reaching its own voter base, which is disproportionately represented among readers of aligned outlets.

To separate favoritism from the government's efforts to reach different audiences, I hold the audience fixed in two ways. First, I restrict attention to advertising campaigns that ran under both administrations with the same stated target audience. If allocation followed the audience a campaign was meant to reach, outlet choice within these campaigns should not have changed with the party in power. Yet aligned outlets remain 11.6 percentage points more likely than misaligned outlets to receive a contract.

Second, I construct two counterfactual benchmarks that show where each campaign's budget would have gone had the government sought only to reach either the campaign's stated demographic or its own voters. Each solves an advertiser's problem, spreading the campaign's observed budget across newspapers at their market advertising prices to reach as many distinct readers as the budget allows. The first is the legal cost-efficiency standard, which requires ministries to purchase the placement that reaches the stated demographic at least cost. The second is the objective an office-seeking politician would

hold (Strömberg, 2004). For each campaign and outlet I take the difference between what the ministry spent and what the benchmark prescribes, and I ask how that gap varies with the outlet's alignment.

Aligned outlets receive $20,876 more per campaign than the legal cost-efficiency standard prescribes and $22,198 more than voter targeting prescribes, and under either benchmark they are approximately 14 percentage points more likely to receive any funding at all. The gap against the cost-efficiency standard amounts to approximately 40 percent of the $51,547 an aligned outlet receives on average per campaign, and it increases with the outlet's alignment. I then quantify the welfare cost of this misallocation in terms of the audience forgone. The same budget could have informed approximately three to four times as many readers.

The reallocation of advertising affects the production of news. I measure productive capacity by the size of the newsroom and by the number of articles an outlet publishes, and I classify those articles by topic into hard news, which covers politics, the economy, and social policy, and soft news, which covers entertainment and sports. I then turn to quality, which concerns the characteristics of an outlet's reporting rather than the volume of its output.

The differential revenue that aligned outlets retain after the transition translates into differential changes in newsroom size and output. Relative to misaligned outlets and to their pre-transition difference, aligned outlets employ 31 percent more affiliated journalists, or approximately 39 reporters when estimated in levels, and publish 50 percent more articles, or approximately 28 articles per day. The differential change is of similar proportion for hard news and soft news, 51 percent and 48 percent, and hard news accounts for approximately 69 percent of the differential change in the articles the outlets write themselves.

I measure quality as the extent to which an outlet's reporting is original rather than outsourced to news-wire agencies, as the speed with which it reaches a story, and as the

amount of verifiable information it contributes to a story. The share of articles obtained from news agencies falls by a statistically insignificant 8 percent, which indicates that the differential change in articles does not come disproportionately from wire copy. I then compare outlets reporting on the same story, grouping articles into news events by textual similarity and holding the event fixed, so that differences reflect how outlets covered a given occurrence rather than which occurrences they faced. Aligned outlets cover 7.8 percentage points more of the events occurring under their own government, and they report 37 percent more unique factual elements per event. The probability that an outlet covers an event on its first day rises by 1.8 percentage points, an estimate that does not separate from zero at conventional levels, and by 2.2 percentage points among hard news events. The coverage and factual-element estimates are statistically significant among hard news events and not among soft news events. Taken together, these results indicate that aligned outlets produce a broader range of content and report news events in greater detail.

Finally, I examine whether the differential change in production and quality reaches readers. Print circulation declined for both groups of outlets between 2014 and 2022, so the difference-in-differences identifies a divergence between their trajectories rather than an absolute increase for either. Circulation at aligned outlets rises 28 percent relative to misaligned outlets, against their pre-transition difference. The Poisson coefficient of 0.246 carries a standard error of 0.165 and does not separate from zero at conventional levels. However, web visits, which do not depend on the circulation figures outlets file with the registry, rise 138 percent at aligned outlets relative to misaligned outlets.

Two alternative explanations could generate that divergence. First, the additional content of higher quality attracts readers, so circulation shifts toward outlets producing more diverse and more detailed reporting. Second, the change in circulation may instead reflect an increase in demand for left-wing media following the 2018 election. To separate them I exploit cross-state variation in the leftward shift in Morena's vote share between 2015 and

2018 and estimate a triple difference-in-differences specification, under the assumption that a state's change in vote share measures the change in its readers' political preferences. An outlet prints one national edition, so a content-driven gain should appear in every state it reaches, whereas a preference-driven gain should concentrate where the electoral shift was largest. The gain concentrates in states with an above-median electoral shift, where the triple interaction is 0.484 with a standard error of 0.230, and the component common to all states is 0.202 and does not separate from zero. The pattern is more consistent with a demand response to the election than with a content-driven gain common to all states, although the estimates are imprecise.

This paper makes three contributions. The first is to the study of government control of the media, which has focused on censorship, propaganda, bribery, and quid pro quo arrangements in autocratic regimes, competitive authoritarian regimes, and backsliding democracies. I document a channel of influence that operates within the two forces that constrain such instruments in democracies, constitutional guarantees of press freedom and market discipline (Besley and Prat, 2006; Gehlbach and Sonin, 2014). In a democracy with a plural press, the government favors aligned outlets and alters the relative strength of ideologically opposed outlets, whether or not any outlet changes what it reports.

The second contribution is to document the consequences of government favoritism in the media market. Most existing work measures the slant of funded outlets (Di Tella and Franceschelli, 2011; Szeidl and Szucs, 2021) or the effect of biased coverage on voting and political attitudes (Adena et al., 2015; Enikolopov et al., 2011). I estimate the consequences of favoritism for market outcomes across competing outlets, namely newsroom capacity, news production, journalistic quality, and circulation. Favoritism is politically attractive because it avoids two costs of censorship, bribery, and explicit quid pro quo arrangements. These instruments are illegal or undemocratic and expose the incumbent to political sanction (Kronick and Marshall, 2024; McMillan and Zoido, 2004), and biased coverage can reduce the audience of aligned outlets (Gehlbach and Sonin, 2014; Durante and Knight,

2012; Knight and Tribin, 2019). Outside closed autocracies, government control of the media coexists with citizen choice, because citizens retain access to independent sources of news (Bor et al., 2026). Favoritism operates within that choice, since it changes the supply of news across competing outlets while citizens remain free to choose among them, and its effects therefore operate at the level of the market.

The third contribution is to the literature on political favoritism in the allocation of public resources. Governments in both democracies and autocracies direct public resources toward the groups they wish to benefit, through patronage and public employment (Padró i Miquel, 2007), transfers to subnational governments (Brollo and Nannicini, 2012), and the geographic distribution of federal spending and grants (Berry et al., 2010; Rosenstiel, 2025). Favoritism is more pronounced where political institutions constrain leaders least, whether that constraint is measured by regime type (Burgess et al., 2015) or by the quality of governance (Hodler and Raschky, 2014). This paper examines favoritism in the market for news. I disentangle favoritism from the objectives that could otherwise explain it by exploiting novel data on the population each campaign states it seeks to reach, a measure that prior studies of government advertising do not observe. Studies of favoritism typically examine which groups receive public resources and, in some cases, how those resources affect the recipients (Hodler and Raschky, 2014). I examine instead how favoritism in public expenditure changes the relative position of competing firms within the market where the government spends.

The remainder of the paper is organized as follows. Section 2 describes the institutional setting for federal advertising in Mexico. Section 3 presents the data and the measurement of outlet ideology. Section 4 documents ideological favoritism in allocation. Section 5 constructs the cost-efficiency and voter-targeting benchmarks. Section 6 traces the downstream effects on newsroom capacity, content production, journalistic quality, and circulation. Section 7 concludes.

2 Institutional Background

I study this mechanism in Mexico because the setting combines two institutional conditions that make favoritism in government advertising politically attractive. First, the federal government is one of the largest advertisers in the national newspaper market and allocates its advertising with considerable discretion, which gives it an instrument to influence the relative strength of competing outlets. Government advertisements account for about a third of the advertisements national newspapers publish, and the median outlet receives 25 percent of its advertising revenue from the federal government (Appendix Figure A3). Second, the national newspaper market is pluralistic and competitive, which raises the cost of direct interference with news content. The largest title holds between 11 and 14 percent of national circulation, so readers can switch to competing outlets when coverage turns biased. Under these conditions, a government that seeks to influence the media gains more from favoring aligned outlets than from interfering with their content.

Federal ministries allocate advertising through a regulated procedure. Each year, every federal ministry submits a communication program to the Ministry of the Interior, which authorizes each campaign, and Congress approves each ministry's advertising budget. For each campaign, the ministry states a communication objective and a target population, and the rules require it to allocate advertising according to each outlet's capacity to reach that population and its advertising rates, a requirement I refer to throughout as the legal cost-efficiency standard. Campaigns must be institutional in character, may not promote any public servant, and must be suspended during electoral campaign periods, except for health, education, and civil protection campaigns. The National Print Media Registry records each outlet's circulation, readership demographics, and advertising rates, so it supplies every input the standard requires.[2]

The national newspaper market is fragmented. During the 1990s, trade liberalization and privatization expanded the private advertising market and reduced the dependence

[2] Appendix A.1 describes the complete regulatory framework.

of newspapers on government subsidies, which allowed investors to launch dailies that relied on commercial revenue (Paxman, 2025, pp. xvii, 65). New titles entered and competed for the same readers, and today eighteen national titles account for 80.9 percent of national print circulation. Over the following decades, however, readers and advertisers moved online, and the private advertising market contracted, most sharply in 2009 (Paxman, 2025). Government advertising therefore became one of the main sources of revenue for national newspapers.

All eighteen titles are privately owned, and most belong to family-owned companies. Only one media conglomerate owns more than one title in the sample, with four of the eighteen titles (Appendix Table B2). The titles are national newspapers, fifteen of the eighteen circulate in at least 28 of the 32 states, and some also publish regional editions. Circulation nonetheless concentrates in Mexico City, which accounts for between 71 and 75 percent of the sample's circulation in each year.

3 Data and Measurement

3.1 Data Sources

I link three datasets, government advertising contracts from federal administrative records, outlet-level variables from the National Print Media Registry, and newspaper content collected from printed editions.

Government advertising contracts. I use administrative records from the Ministry of the Interior's Social Communication Strategy program, which cover every federal advertising contract signed between 2014 and 2022. Each contract identifies the ministry, the recipient outlet, the date, the amount spent, the number of advertisements by size, and the price paid per advertisement. I complement these records by collecting a novel dataset on campaign characteristics, which reports the communication objective, the target demographic, and whether the campaign promotes government policies and achievements,

social programs, prevention and awareness, or civic and cultural obligations.[3]

Outlet-level variables. I construct a novel panel dataset on time-varying outlet characteristics from the National Print Media Registry, which every outlet seeking federal advertising must update each year. For each outlet and year the panel reports average daily circulation by state, the share of readers by socioeconomic level, gender, education, and age group, the legal name of the owner, and list prices per cm^2 for advertising space across sections and page locations, which I average into a single market price per outlet. I restrict the sample to national outlets that publish news daily (or at least five days per week), dropping magazines and outlets focused only on entertainment or sports. This leaves approximately 25 national outlets per year. I further limit the analysis to 18 outlets for which complete data on daily articles and historical registry information are available. On average, these 18 outlets account for 80.9 percent of national print circulation each year.

News content. I collect the text of all print issues for the 18 national outlets from 2014 to 2022, totaling approximately 6 million articles.[4] From the text I measure the number of articles an outlet publishes, its split between hard and soft news, and the news events on which I compare coverage across outlets. From bylines I measure newsroom size, and I separate wire copy (Reuters, Associated Press, Notimex) from house staff and named journalists so that the measure counts an outlet's own reporters. I classify every printed advertisement as public or private. The classification validates the advertising spending ministries report and measures the share of an outlet's advertising space that the government occupies against the share that private advertisers purchase.

Table 1 reports summary statistics for the full sample, separately for the right-wing (PRI) and left-wing (Morena) administrations. Government advertising falls from $1.7 million per outlet under PRI to $467,000 under Morena, a decline that, if distributed un-

[3] I obtain these characteristics through information requests to federal ministries under the National Transparency Platform.

[4] See Table B1 for per-outlet totals and yearly averages.

evenly across outlet types, translates into revenue shocks that differ by outlet ideology. Even at these levels, government advertising accounts for roughly one-third of all advertising space in the average outlet, with a standard deviation of 17 percentage points indicating that some outlets depend far more heavily on public contracts than others. Ministries also pay above the market price, \$21.93 per cm^2 against a market price of \$7.83 under PRI and \$13.41 against \$7.52 under Morena, or 2.8 and 1.8 times the market price. The average outlet holds a 5.6% market share, consistent with a market where no outlet dominates. Together, a government advertiser that accounts for one-third of advertising space and prices that exceed market rates describe a setting in which government choices can reshape the market.

Table 1: **Summary Statistics**

	Right-Wing Government (PRI) (2014–2018)		**Left-Wing Government (Morena)** (2018–2022)	
	Mean	SD	Mean	SD
Panel A: Government Advertising				
Gov. advertising per outlet (USD, annual)	1,709,158	1,769,201	466,711	826,105
Price paid by government (USD per cm^2)	21.93	16.90	13.41	12.10
Share of public advertising (%)	34.1	17.0	33.8	16.9
Number of ministries contracting	11.0	5.9	6.0	3.8
Panel B: Outlet Characteristics				
Daily circulation (thousands)	115.1	56.6	115.7	57.8
Average market share (%)	5.7	2.8	5.6	2.7
Monthly articles	2,497	1,625	2,275	1,273
Newsroom size	198	160	204	139
Market advertising price (USD per cm^2)	7.83	4.16	7.52	3.42

Outlet-year observations. The sample covers 18 national newspapers from 2014 to 2022. USD values in 2022 constant dollars. Gov. advertising per outlet aggregates all federal ministry contracts per outlet and year. Share of public advertising is the fraction of observed advertisements classified as government-sourced from newspaper content. Market advertising price is the average list price per cm^2.

3.2 Measuring Ideological Alignment

Identifying ideological favoritism requires a measure of where each outlet stands on the left-right dimension. I construct one from text, following the logic of Gentzkow and Shapiro (2010) that the differential use of partisan language identifies ideological position,

and using the penalized estimation to recover a sparse and interpretable set of partisan phrases (Gentzkow et al. (2019)). I estimate the measure using editorial board pieces since they reflect the outlet's institutional stance.

Fifteen of the eighteen outlets publish a dedicated editorial section, and I use the 8,938 editorials they published between 2014 and 2017. I end the estimation window before the transition for two reasons. The measure never sees vocabulary specific to the incoming administration, so the recovered dimension separates left from right and not support for the 2018 winner from opposition to it. Outlet ideology also becomes a pre-treatment characteristic, fixed before any of the advertising reallocation I study.[5]

One outlet serve as left-wing anchor, La Jornada, and two as right-wing anchors, Reforma, and El Economista. I select these three based on their editorial histories and established positions in the Mexican media landscape. La Jornada is the most left-wing Mexico City daily in the ideological scores of Lawson (2002) and one of the country's most prominent left-wing outlets (Paxman, 2025), and Reforma lies to the right of center in the same scores (Lawson, 2002). The estimation vocabulary is built from the 3,836 editorials these three outlets published in the window, subject to three restrictions, a phrase must appear in at least three of the three anchor outlets, so that no single outlet's recurring vocabulary can define the dimension; it must appear in at least three anchor editorials; and it must not contain an outlet name, a column title, or a formatting artifact.

I represent each anchor editorial as a vector of phrase proportions that sums to one, and I fit an L_1-penalized logistic regression of an indicator for right-wing anchor authorship on that vector. The penalty is selected by scanning sixty values of the regularization parameter and retaining the solution closest to a target of five hundred non-zero phrases, following the sparsity target in Gentzkow et al. (2019). The selected model retains 514 phrases, 224 of which load right-wing and 290 left-wing. Table 2 reports a subset cho-

[5]Phrases are the unigrams and bigrams formed from the stemmed text. I lowercase each editorial, strip accents, digits, and punctuation, remove Spanish stopwords, lemmatize, and stem, dropping tokens shorter than three characters.

sen for interpretability. Phrases loading left-wing include pension, Cuba, promise, and countryman, while phrases loading right-wing include investment, institution, economic growth, and competence. The two sets divide social provision and national identity from markets and technocratic competence, which is the axis on which Mexican parties have divided since the 1990s.

Table 2: **Top Partisan Phrases**

Right-wing Phrases			
American (estadounidens)	Capacity (capac)	Official (funcionari)	Local (local)
Competence (competent)	Union (sindicat)	Investment (inversion)	Structure (estructur)
Professor (profesor)	Institution (institucion)	Org. crime (crim organiz)	Internet (internet)
Transformation (transform)	Corruption (corrupcion)	Citizenship (ciudadani)	Process (proces)
State-run (estatal)	Service (servici)	Opportunity (oportun)	Infrastructure (infraestructur)
Freedom (libert)	Risk (riesg)	Institutional (institucional)	Poll (encuest)
Criminal (criminal)	Growth (crecimient)	Fiscal (fiscal)	
Left-wing Phrases			
Tax auth. (sat)	Injustice (injust)	Environmental (ambiental)	Hunger (hambr)
War (guerr)	Ministry (secretari)	Gas hike (gasolinaz)	Common sense (sent comun)
People (puebl)	Att. General (procurador)	Pollution (contamin)	Teacher (maestr)
Electoral roll (padron)	Promise (promes)	Countryman (paisan)	Disarmament (desarm)
Government (gobern)	Gov. official (mandatari)	Odebrecht (odebrecht)	Misery (miseri)
Ayotzinapa (ayotzinap)	Femicide (feminicidi)	Health (salud)	Electoral process (proces electoral)
Pension (pension)	Cuba (cub)	North American (norteamerican)	

The L_1-penalized logistic regression retains 514 phrases with non-zero loadings, 224 loading right-wing and 290 loading left-wing. This table reports 55 of them, selected for interpretability and not by the rank of the loading. Phrases are shown in English with the Spanish stem in parentheses.

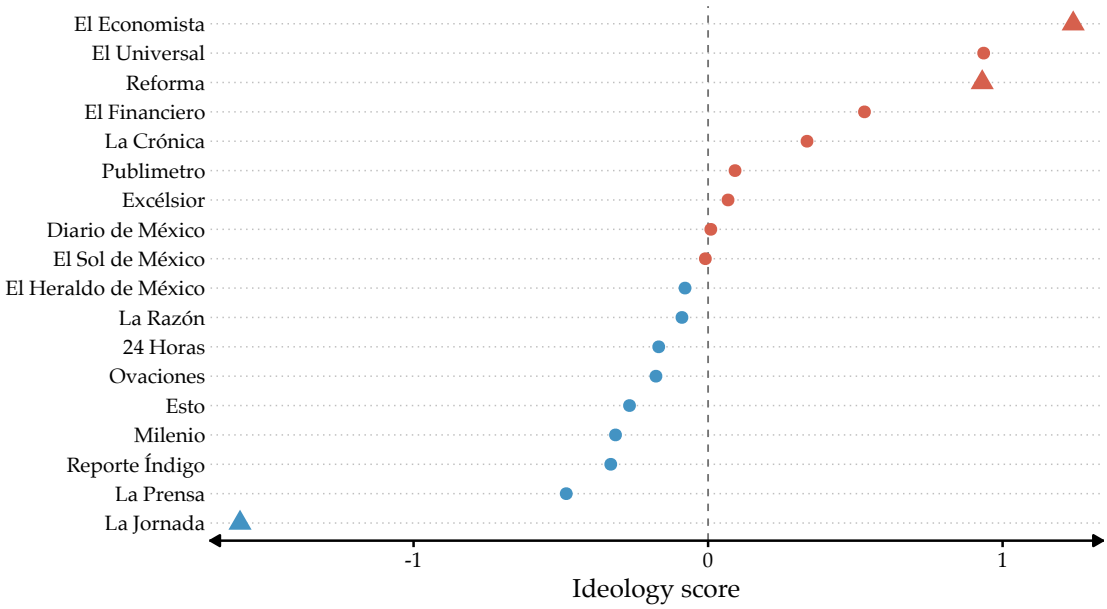
Each article receives a score equal to the inner product of the estimated loadings with its own vector of phrase proportions, and each outlet's score is the mean of that quantity across its articles. I subtract the midpoint between the left- and right-wing anchor group means, so that zero falls equidistant between the two clusters, and I classify an outlet as left-wing when its recentered score is negative and right-wing otherwise. The three anchors are scored by the same rule as every other outlet and are not held at their training labels.

Diario de México, El Heraldo de México, and Esto publish no editorial section, so I score them by applying the estimated loadings to their news articles under the same preprocessing, scoring rule, and midpoint. Appendix C.1.1 resamples each of these three

corpora down to the size of a typical editorial corpus and reports the resulting distribution of scores, which is stable in every draw.

Figure 1 plots each outlet's recentered score, with La Jornada, Reforma, and El Economista marked as triangles. Nine outlets fall on each side. Several score close to the midpoint, among them the four with the thinnest editorial corpora. I treat each outlet's ideology as constant over the sample period. Appendix Figure C1 plots the average editorial slant of left-wing and right-wing outlets in each year from 2014 to 2022. The two groups remain clearly separated across the full period and their ordering never reverses, indicating that the classification estimated on the pre-transition window is not specific to that window.

Figure 1: **Outlet Ideology Scores**



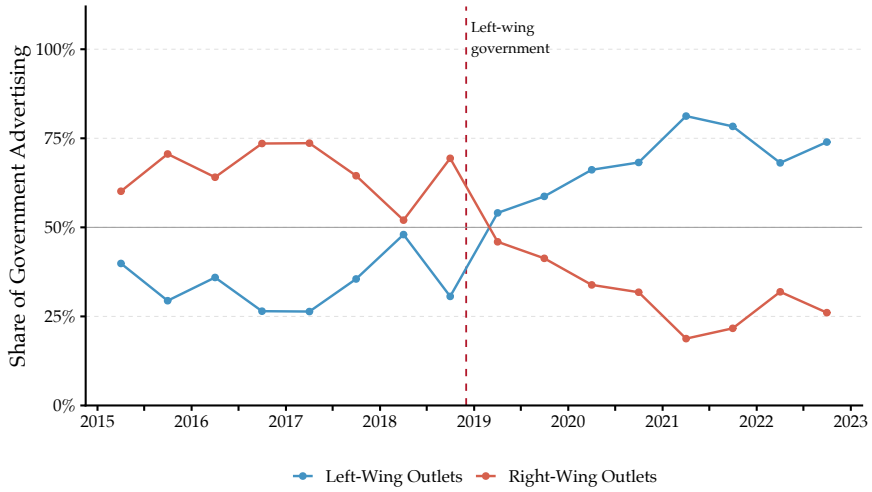
Phrase loadings are estimated by L_1-penalized logistic regression on the editorials of the three anchor outlets. Triangles mark the anchors. Negative values indicate left-leaning positions and positive values indicate right-leaning positions.

3.3 Motivating Patterns

Before turning to the causal analysis, I document descriptive patterns in government advertising allocation. Figure 2 plots, for each semester, the share of federal advertising

spending on the 18 sample newspapers that goes to left-wing and to right-wing outlets.

Figure 2: **Share of Government Advertising by Outlet Ideology**



Share of federal ministry advertising spending on the 18 sample newspapers received by left-wing outlets (9) and right-wing outlets (9) in each semester. The dashed line marks the December 2018 presidential transition from PRI to Morena.

Under the right-wing PRI administration, right-wing outlets received the larger share of government advertising in every semester, between about one half and three quarters of the total. Following the December 2018 transition to Morena, the pattern reverses. The share of left-wing outlets exceeds one half in every semester and reaches about four fifths by 2021. Shares abstract from the level of spending, which differs sharply across administrations. Spending fell for both groups of outlets after the transition, consistent with the austerity the incoming government announced, and stabilized at the lower levels shown in Table 1. Appendix Figure A4 plots average spending per outlet in levels.

The raw patterns are consistent with ideological favoritism but do not establish it. Three explanations would produce the same reversal without any preference for aligned outlets. The incoming government has different policy priorities and therefore advertises different programs. Those programs target different audiences, and if left-wing outlets reach the audiences a left-wing government cares about, the reallocation reflects whom each campaign set out to inform. The government may also seek to direct the benefits

of its campaigns toward particular audiences, such as its own voters, whose choice of newspaper may correlate with an outlet's alignment with the government. Sections 4 and 5 identify the effect of ideological alignment on allocation and address each of these explanations in turn.

4 Favoritism in Advertising Allocation

Governments may direct advertising toward outlets that share their ideology. To test whether such favoritism shapes allocation decisions, I examine whether a change in government shifts spending toward ideologically aligned outlets. This section also addresses a change in the mix of campaigns as an alternative explanation, and Section 5 addresses the pursuit of particular audiences.

I employ a difference-in-differences specification that leverages variation in government transitions and the ideological alignment of newspapers. I focus on the presidential transition in 2018, where the left-wing party Morena came to power, replacing the right-wing party PRI. Newspaper ideological positions are fixed over time and inferred from the text-based measures described in Section 3.2. I estimate the following regression:

$$Y_{ngt} = \beta \text{ Left-Wing Gov}_t \times \text{ Left-Wing Outlet}_n + \mu_t + \delta_{ng} + \epsilon_{ngt}, \tag{1}$$

where Y_{ngt} represents one of two outcome variables of government advertising to outlet n from ministry g in year t. First, an indicator for whether the outlet receives any advertising contract (extensive margin). Second, the level of total advertising spending in 2022 USD, including zero values for outlet–ministry pairs with no contracts in a given year (intensive margin). The price paid, conditional on a contract, is a third margin the government could in principle use, but it is observed only where a contract was placed and alignment moves that margin, so it is not point identified by this design; Appendix E.1 reports it with bounds. The variable Left-Wing Gov_t is an indicator equal to 1 when the

presidency is held by a left-wing party, and Left-Wing Outlet$_n$ equals 1 if the outlet is classified as left-leaning and 0 if right-leaning.

Without ministry fixed effects, the estimates could reflect cross-ministry variation in budget trajectories, a ministry that structurally favors left-wing outlets for non-ideological reasons (e.g., because its target audience skews left) may also have expanded its budget under the left-wing government, generating a spurious correlation. δ_{ng} removes this confound by identifying β from within-pair changes. Hence, the panel is structured at the outlet × ministry × year level. The specification also includes year fixed effects (μ_t) to absorb shocks common to all outlets in a given year, including the aggregate path of the federal communication budget. Standard errors are clustered at the newspaper level, as the treatment variable is defined at the newspaper level and errors across all ministry relationships of the same newspaper are correlated by construction. The identification strategy relies on the common trends assumption, in the absence of a shift in the party in power, advertising allocations to left- and right-leaning outlets would have followed common trends.

Table 3 presents the results. Columns (1) and (2) use the full sample of all advertising campaigns, restricted to the 10 ministries that ran campaigns across both administrations.[6] These columns report the extensive margin and the intensive margin in levels.

Ideological alignment favors aligned outlets on both the extensive and the intensive margin. On the extensive margin, a left-wing outlet is 18.3 percentage points more likely than a right-wing outlet to receive a contract from a given ministry once the left-wing government takes office. As on the intensive margin, this is a differential and not a gain in levels. The contract rate fell from 0.56 to 0.41 for aligned outlets and from 0.78 to 0.44 for misaligned ones. The intensive margin requires more care, because the incoming administration cut advertising across these ten ministries by 48 percent and both groups of outlets lost revenue. Spending on aligned outlets fell by $16,096 per ministry-year, or 15 percent

[6]The 10 ministries are Culture, Defense, Education, Energy, Foreign Affairs, Health, Interior, Navy, Presidency, and Tourism.

of their pre-transition average, while spending on misaligned outlets fell by $115,994, or 72 percent. Those two declines differ by $99,898; the coefficient of $105,362 is the same comparison net of outlet-ministry and year fixed effects, approximately 0.33 standard deviations of the outcome. Alignment therefore protected an outlet from a contraction that fell almost entirely on its competitors, and did not raise what any outlet received. The levels estimate in Column (2) averages over pairs that begin receiving advertising after the transition and pairs that received advertising under both administrations, so it captures both margins at once. Because treatment varies across only 18 outlets, the table also reports wild cluster bootstrap and randomization inference p-values. Both are close to the clustered p-values, at 0.041 and 0.047 for the extensive margin and 0.044 and 0.048 for the intensive margin.

Table 3: **Differential Effects of Government Partisanship on Advertising Spending Across Left- and Right-Wing Outlets**

	All campaigns		**Campaigns run under both governments**	
	1{Adv > 0}	Spending	**1**{Adv > 0}	Spending
	(1)	(2)	(3)	(4)
Left-Wing Government × Left-Wing Outlet	0.183**	105,362*	0.116**	9,360*
	(0.082)	(53,749)	(0.042)	(5,311)
Wild cluster bootstrap p	[0.041]	[0.044]	[0.012]	[0.072]
Randomization inference p	[0.047]	[0.048]	[0.021]	[0.064]
Pre-2018 mean, left-wing outlets	0.56	107,562	0.19	4,449
Pre-2018 mean, right-wing outlets	0.78	160,559	0.29	9,696
Year FE	✓	✓	✓	✓
Outlet × Ministry FE	✓	✓		
Campaign × Outlet FE			✓	✓
Observations	1,410	1,410	2,397	2,397

Note: *p<0.1; **p<0.05; ***p<0.01. SEs clustered at the outlet level in parentheses. With 18 clusters, brackets report wild cluster bootstrap p-values (null imposed, Rademacher weights, 1,999 draws) and randomization inference p-values from 5,000 random reassignments of the left-wing label across the 18 outlets, holding the number of left-wing outlets fixed. Cols. (1) and (3) use an indicator for positive advertising spending (extensive margin). Cols. (2) and (4) use the level of advertising spending in 2022 USD, including zeros. Cols. (1)–(2) use the outlet–ministry–year panel for the 10 ministries with campaigns run under both governments (N = 1,410). Cols. (3)–(4) use the campaign–outlet–year panel for the 17 campaigns implemented across both administrations with identical target demographics (N = 2,397). The price paid per unit of space is reported in Appendix Table E3. Means computed over the pre-2018 period.

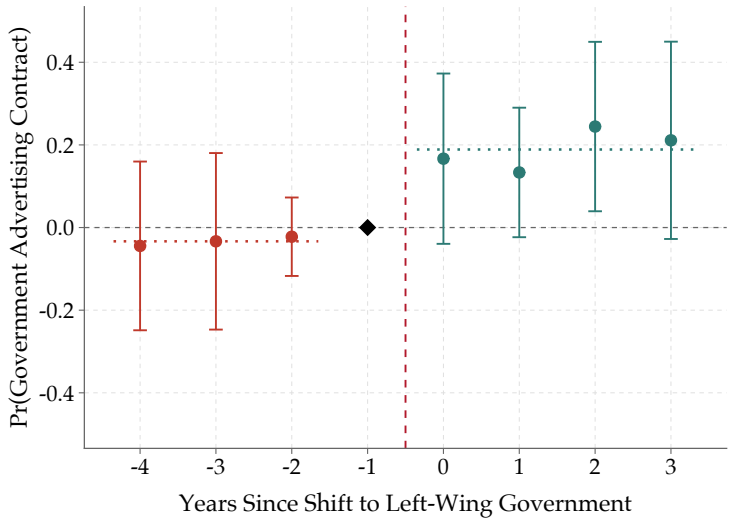
Higher spending on aligned outlets could reflect ideological favoritism, or it could reflect a change in policy priorities that shifts spending toward campaigns typically advertised in aligned outlets. A left-wing government, for example, may place greater emphasis on social policy and shift spending toward social policy campaigns, which may be advertised more heavily in left-wing outlets. To separate these explanations, Columns (3) and (4) restrict the sample to the 17 campaigns that ran under both governments with identical stated target demographics and include campaign × outlet fixed effects. The coefficient therefore compares the same outlet carrying the same campaign, addressed to the same audience, before and after the transition, so a change in the mix of campaigns or in the audience the government sought to reach cannot account for it.

Favoritism persists within these campaigns. Aligned outlets are 11.6 percentage points more likely than misaligned outlets to receive advertising for a given campaign, with wild cluster bootstrap and randomization inference p-values of 0.012 and 0.021, and they receive \$9,360 more per campaign and year, an estimate that clears the ten percent level. Contract rates fall for both groups, from 0.19 to 0.13 for aligned outlets and from 0.29 to 0.12 for misaligned ones. Within these campaigns, however, aligned outlets gain in levels, with spending rising from \$4,449 to \$8,382 per campaign and year, while spending on misaligned outlets falls from \$9,696 to \$5,032. The reallocation toward aligned outlets therefore does not reflect a change in the mix of campaigns.

Favoritism therefore operates through the choice of outlets and the amount spent, and not through the price paid per unit of space. Holding the advertisement's size and the campaign fixed, the price paid to an aligned outlet is statistically indistinguishable from the price paid to a misaligned one, and the bounds that account for selection into a contract contain zero, as Appendix E.1 reports. Appendix D.1 provides two further exercises following Chen and Roth (2024), namely a Poisson regression whose coefficient is scale-invariant and reads as a semi-elasticity, and Lee bounds (Lee, 2009) for the conditional intensive margin.

To assess the validity of the parallel trends assumption, Figure 3 presents event study estimates corresponding to Column (1) of Table 3. Event time is defined as 12-month periods relative to the December 2018 transition, with the period immediately preceding the transition (December 2017–November 2018) normalized to zero. Following the transition, the coefficients become positive and statistically significant, indicating a sharp increase in advertising spending to left-wing outlets relative to right-wing outlets, and the effect grows persistently through 2022.

Figure 3: **Event Study of the Extensive Margin of Government Advertising by Outlet Ideology**



Coefficients on the interaction between event time and a left-wing outlet indicator, from Column (1) of Table 3. Reference group: right-wing outlets. Omitted period: December 2017–November 2018. Error bars show 95% confidence intervals; standard errors clustered at the outlet level. Outlet–ministry fixed effects included.

The flat pre-transition coefficients and the increasing post-transition estimates are consistent with a discontinuous change in advertising allocations driven by the party transition and not by pre-existing differences in trajectory. Event study plots for other specifications are presented in Appendix Figure D1.

5 Separating Favoritism from Audience Reach

Section 4 documents that ministries shifted advertising toward aligned outlets after the transition, including within campaigns that ran under both governments. The same pattern could arise without favoritism if the government redirected the benefits of its campaigns toward particular groups, such as its own voters, and those groups read aligned outlets. This section separates favoritism from that alternative. For each campaign, I construct the allocation a ministry would have chosen had it sought only to reach a given audience with its budget, and I compare the observed allocation with that benchmark. Because the benchmark solves the allocation of each campaign's budget across all outlets jointly, it also quantifies the distortion that favoritism introduces, measured in dollars per outlet and in the readers each campaign could have informed with the same budget.

5.1 The Advertiser's Problem

A ministry runs campaign j in year t with budget B_{jt}. Outlet i sells an insertion at price p_{it} and reaches a_{ijt} readers in the campaign's target audience, so spending s_{ijt} buys $x_{ijt} = s_{ijt}/p_{it}$ insertions.[7] The ministry chooses spending to maximize the number of distinct target readers it informs,

$$\max_{\{s_{ijt} \geq 0\}} \sum_i a_{ijt} \left[1 - (1-\phi)^{x_{ijt}}\right] \qquad \text{s.t.} \qquad \sum_i s_{ijt} \leq B_{jt} \tag{2}$$

where ϕ is the probability that a reader observes any one insertion, so the bracketed term is the probability that a reader of outlet i observes the campaign at least once.[8]

The problem rests on three assumptions. First, exposures are independent across

[7] Outlets set their advertising rates at the beginning of each year and report them to the National Print Media Registry, so prices are fixed when ministries allocate their campaign budgets.

[8] Equation 2 is the reach-only case of Gentzkow et al. (2024), who value an advertiser reaching a viewer M times at $a \cdot u(M)$ for a submodular $u(\cdot)$ and set $u(M) = \mathbf{1}[M > 0]$. It also inherits the constant-reach, independent-readership technology of Grossman and Shapiro (1984), who write the reach of a campaign placed in m vehicles each covering a fraction r of the population as $1 - (1-r)^m$.

insertions. Second, only the first exposure carries value, as in campaigns that inform rather than persuade. The x-th insertion in an outlet therefore informs only the share $\phi(1-\phi)^{x-1}$ of its target readers who did not observe earlier insertions, so ϕ determines the rate at which returns to insertions diminish. Higher values of ϕ make returns diminish faster and spread the benchmark allocation across more outlets, while lower values concentrate it in fewer outlets. Third, readers do not multi-home across the sample newspapers. Benchmark campaigns fund more outlets than observed campaigns, so multi-homing would reduce benchmark reach more than observed reach, and the reach gap I report is therefore an upper bound. Because reach within an outlet saturates at a_{ijt}, a ministry that maximizes reach spreads its budget across outlets.

5.2 Constructing the Benchmarks

The problem has a closed-form solution. The objective is strictly concave and additively separable, so the Kuhn–Tucker conditions yield a unique optimum at which the budget binds. With λ_{jt} denoting the multiplier on the budget constraint and $\kappa \equiv -\ln(1-\phi)$,

$$s^*_{ijt}(\lambda_{jt}) \;=\; p_{it} \cdot \max\left\{0,\; \frac{1}{\kappa}\ln\left(\frac{a_{ijt}\,\kappa}{\lambda_{jt}\,p_{it}}\right)\right\} \tag{3}$$

Total benchmark spending decreases strictly in λ_{jt}, so exactly one value of λ_{jt} exhausts each campaign's budget, and I compute it numerically for each of the 383 campaigns. The max operator generates an extensive margin. An outlet whose cost per target reader, p_{it}/a_{ijt}, exceeds κ/λ_{jt} receives no spending, so the benchmark determines which outlets a ministry funds as well as the amount each receives. I set $\phi = 0.10$ since the advertising data do not identify ϕ, Appendix E.3 re-estimates every deviation over a range of values. The intensive-margin deviation is positive throughout, and the continuous-score estimate is significant at each value.

The two benchmarks differ only in the audience a_{ijt}. The *cost-efficiency benchmark* sets a_{ijt}

to an outlet's readers in the campaign's stated demographic, the audience the legal cost-efficiency standard requires ministries to reach.[9] The *voter-targeting benchmark* sets a_{ijt} to an outlet's readers who support the incumbent, predicted by weighting official state-level vote shares by the outlet's circulation across states. This measure correlates at 0.80 with an independent estimate constructed from individual polling microdata (Appendix E.2).

For each benchmark, I regress the difference between observed and benchmark spending on alignment,

$$Y_{ijt} = \alpha \cdot \mathbf{1}[\text{Aligned}]_{it} + \gamma_j + \delta_i + \varepsilon_{ijt} \tag{4}$$

where γ_j and δ_i are campaign and outlet fixed effects and $\mathbf{1}[\text{Aligned}]_{it}$ equals one when the outlet shares the ideology of the party running the campaign. The outcome is either the extensive margin, $\mathbf{1}[s^{\text{obs}}_{ijt} > 0] - \mathbf{1}[s^{*}_{ijt} > 0]$, or the intensive margin, $s^{\text{obs}}_{ijt} - s^{*}_{ijt}$ in dollars. Campaign fixed effects absorb the budget, the stated demographic, and the party in power, so α compares outlets within a campaign. Standard errors are clustered by outlet, and each specification is re-estimated with the continuous ideology score in place of the binary indicator.

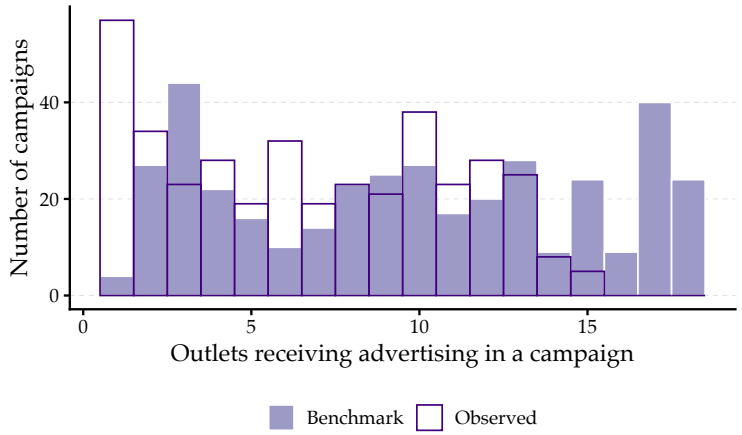
5.3 Deviations from the Benchmarks

I first present results on the extensive margin, which records whether an outlet receives government advertising at all. Figure 4 plots how many outlets ministries funded per campaign and how many the benchmark prescribes. The observed distribution has a mode of one outlet and a median of six, while the benchmark distribution has a median of eleven. Ministries concentrate their campaigns in fewer outlets than reaching the stated demographic would require. The number of outlets a benchmark campaign funds rises

[9] Campaigns state a demographic on two dimensions, income and age, and the registry reports each outlet's readership shares on the same categories. I compute the share of an outlet's readers in the stated groups on each dimension and multiply the two shares. This treats income and age as independent within an outlet, because the registry reports only the marginal distributions. I drop three of the 386 campaigns, whose stated demographic no national newspaper reaches. The estimation sample holds 383 campaigns and 6,711 outlet–campaign cells.

with ϕ, from one as ϕ approaches zero to 10.3 at $\phi = 0.10$ and 15.4 at the upper end of the range, so this comparison depends on the calibration of ϕ (Appendix E.3).

Figure 4: **Outlets Used per Campaign, Observed and Benchmark**



The benchmark solves equation 2 at $\phi = 0.10$ for each campaign, holding the observed budget fixed. Bars overlap where the distributions coincide.

Table 4 reports estimates of equation 4 for both outcomes. On average, a campaign funds 44 percent of aligned outlets and 33 percent of misaligned outlets. In Panel A, aligned outlets are 13.9 percentage points more likely than misaligned outlets to receive advertising that the cost-efficiency benchmark does not allocate to them, and 13.1 percentage points more likely under the continuous ideology score. Columns (3) and (4) repeat the comparison against the voter-targeting benchmark, and both estimates are larger. Alignment therefore predicts which outlets a ministry funds beyond what either benchmark prescribes.

I then present results on the intensive margin, which records how much a funded outlet receives. An aligned outlet receives $51,547 per campaign on average, against $28,833 for a misaligned outlet. In Panel B, aligned outlets receive $20,876 more per campaign than the cost-efficiency benchmark prescribes, or 40.5 percent of the average for aligned outlets, and $28,749 more under the continuous score. Against the voter-targeting benchmark, the binary deviation is $22,198, or 43.1 percent of that average. The deviation is positive and significant for both alignment measures and both benchmarks.

Table 4: **Deviations from the Benchmark Allocations by Outlet Alignment**

	Cost-efficiency benchmark		Core-supporter benchmark	
	(1)	(2)	(3)	(4)
Panel A. Extensive margin				
Alignment	0.139***	0.131***	0.141***	0.153***
	(0.039)	(0.018)	(0.048)	(0.022)
Observed mean, aligned	0.44	0.44	0.44	0.44
Observed mean, misaligned	0.33	0.33	0.33	0.33
Panel B. Intensive margin, 2022 dollars				
Alignment	20,876**	28,749***	22,198**	31,038***
	(8,912)	(6,008)	(9,865)	(6,863)
Observed mean, aligned	51,547	51,547	51,547	51,547
Observed mean, misaligned	28,833	28,833	28,833	28,833
Deviation as share of aligned mean	40.5%		43.1%	
Alignment measure	**Binary**	**Continuous**	**Binary**	**Continuous**
Ad Campaign FE	✓	✓	✓	✓
Outlet FE	✓	✓	✓	✓
Observations	6,711	6,711	6,711	6,711

Estimates of equation 4 at $\phi = 0.10$. Panel A takes $\mathbf{1}[\text{observed} > 0] - \mathbf{1}[\text{benchmark} > 0]$ as the outcome, and Panel B takes the difference in 2022 dollars. Each column reports a separate regression in which the stated alignment measure is the only regressor. Binary columns split outlets at the ideological midpoint. Continuous columns replace the split with the ideology score of Section 3.2, signed so that higher values indicate closer alignment with the incumbent. Binary and continuous coefficients are measured on different scales and are not comparable in magnitude. Observed means are computed over outlet–campaign cells by alignment, and the share row divides each binary deviation by the observed mean for aligned outlets. Standard errors clustered by outlet across 18 clusters. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.10$.

The copartisan measure weights state vote shares by where an outlet circulates, so it does not capture the sorting of readers within a state. Appendix E.4 rebuilds the voter-targeting benchmark under assumed levels of sorting. Measured in dollars, the continuous-score deviation remains significant at the five percent level as long as the odds that an incumbent supporter reads an aligned rather than a misaligned newspaper stay within 2.1 times the measured odds, and at the ten percent level within 2.7 times. Sorting of that magnitude is unlikely for two reasons. The two independently constructed copartisan measures differ between left-wing and right-wing outlets by at most 0.4 percentage points (Appendix E.2). In addition, every weighting of reader demographics predicts the within-

campaign allocation with the wrong sign, while a single parameter that loads circulation on outlet ideology predicts it with the right sign (Appendix E.5).

Finally, I quantify the cost of the deviation from the cost-efficiency benchmark in readers informed. Campaigns inform between 178,000 and 275,000 target readers, against the 765,000 that the benchmark informs with the same budgets, or 23 to 36 percent.[10] Every campaign informs fewer readers than its budget could have informed (Appendix Figure E1). The shortfall reflects the choice of outlets and not the prices paid, because both allocations convert spending into insertions at the same market price (Appendix E.1).

Aligned outlets receive more advertising than a ministry would allocate to them if it sought only to reach the campaign's stated demographic or the incumbent's voters, and the gap increases with an outlet's alignment. The government therefore favors outlets that share its ideology.

6 Downstream Effects in the Media Market

This section documents how favoritism in the allocation of advertising can become a form of media capture, in which the government gains influence over the media by altering the relative strength of competing political voices. Federal advertising accounts for 25 percent of the advertising revenue of the median outlet between 2014 and 2018 (Appendix Figure A3), so a change in the party in power changes the probability that an outlet receives a substantial share of its revenue. I estimate the consequences of the transition with the difference-in-differences design of Section 4, which compares left-wing and right-wing outlets before and after the transition and includes outlet and time fixed effects. The market-level outcomes include newsroom capacity, news production, journalistic quality, and circulation.

[10] The estimate of 178,000 readers uses the number of advertisements each ministry records. The estimate of 275,000 readers converts each ministry's spending into advertisements at the private market price (Appendix E.1).

The 2018 election also produced a large political realignment, as López Obrador won the presidency with 53 percent of the vote. If readers who turned toward Morena also turned toward left-wing outlets, the difference-in-differences would attribute a shift in demand to the reallocation of advertising. A contemporaneous shock to private advertising at left-wing outlets would have the same effect. I therefore examine the timing of each response relative to the December 2018 inauguration, the response of private advertising, and, for circulation, cross-state variation in the electoral shift.

6.1 Newsroom Capacity, News Production, and Quality

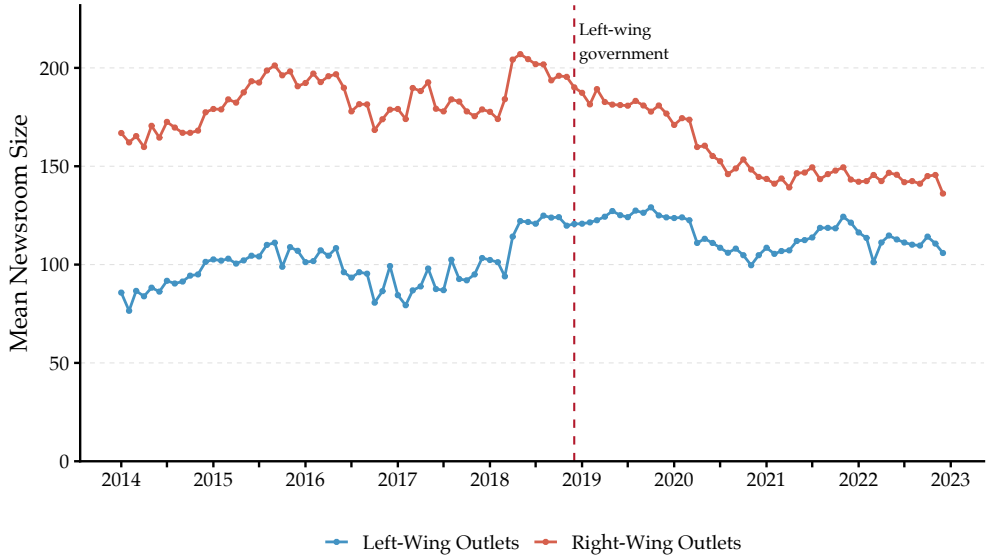
Advertising revenue finances newsroom staff, so a decline in revenue reduces the number of journalists an outlet employs and the number of articles it publishes. I measure three outcomes, the size of the newsroom, the number of articles, and the quality of reporting.

Figure 5 plots the average number of affiliated journalists per outlet in each month from 2014 to 2022.[11] Right-wing outlets have larger newsrooms throughout the sample. The gap narrows during 2018, before the December transition, as the left-wing series rises from approximately 100 to 120 journalists. It narrows further after the transition, as right-wing newsrooms decline from approximately 190 to 140 journalists while left-wing newsrooms remain between approximately 105 and 125. The narrowing during 2018 may reflect readers who turned toward left-wing outlets after the July election. Ministries reallocated advertising only after the new administration took office, so the revenue shift dates from the end of 2018. Because the raw series cannot separate the revenue shift from other shocks common to both groups, I estimate the difference-in-differences specification.

Table 5 estimates equation 1 on the outlet–month panel by Poisson pseudo-maximum likelihood, so that $\exp(\hat{\beta}) - 1$ measures the proportional change in the mean count (Chen and Roth, 2024). The outcome in Column (1) is newsroom size, the number of affiliated

[11] A journalist is active in a month if at least one article carries their byline, excluding agency copy. The sample includes authors with at least 12 articles who publish at least 75 percent of their articles in one outlet.

Figure 5: **Newsroom Size by Outlet Ideology, 2014–2022**



Newsroom size is the number of affiliated journalists active per outlet-month. Affiliated journalists are bylined authors with at least 12 articles who publish at least 75 percent of their articles in one outlet. Journalists are identified through byline analysis, excluding agency content.

journalists who publish under their own byline in a given month. Affiliated journalists write exclusively or predominantly for one outlet. A count of all bylines would increase mechanically with output, because occasional contributors would enter the monthly count. Appendix F.3.2 reports that count and the classification of journalists. Columns (2) to (4) use total articles and the number of articles classified as hard or soft news, excluding wire copy. The classification comes from a topic model described in Appendix F.1. Column (5) uses the share of articles obtained from news wires, estimated on wire articles with the logarithm of total articles as an offset.

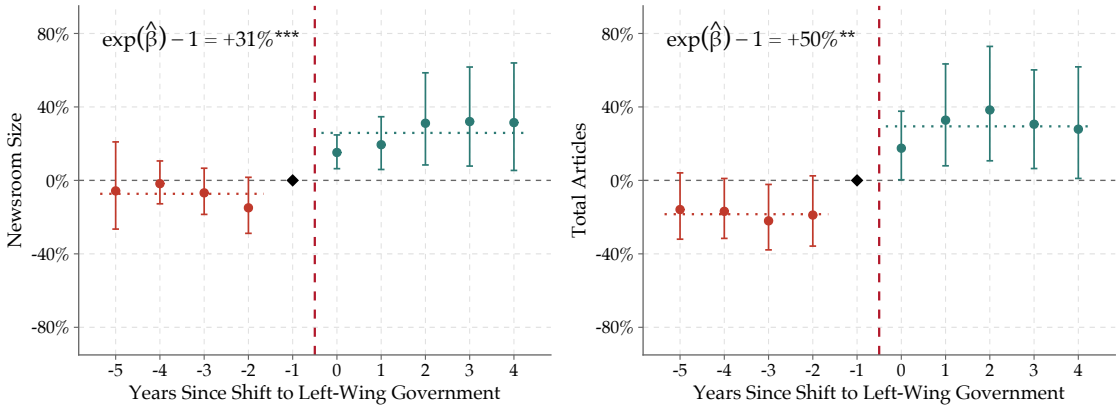
Table 5: **Differential Effects of Government Partisanship on Content Production Across Left- and Right-Wing Outlets**

	Newsroom (1)	Total Articles (2)	Hard News (3)	Soft News (4)	Wire Share (5)
Left-Wing Gov × Left-Wing Outlet	0.273***	0.405**	0.411***	0.392**	−0.085
	(0.087)	(0.143)	(0.107)	(0.172)	(0.178)
$\exp(\hat{\beta}) - 1$	31%	50%	51%	48%	-8%
Mean pre-2018 (Left)	99.5	1,540.4	855.4	459.0	14.7%
Mean pre-2018 (Right)	183.7	2,700.8	1,566.7	740.3	14.6%
Year-Month FE	✓	✓	✓	✓	✓
Outlet FE	✓	✓	✓	✓	✓
Observations	1,908	1,908	1,908	1,908	1,899

Note: *p<0.1; **p<0.05; ***p<0.01. SEs clustered at the outlet level, with p-values from a t distribution with 17 degrees of freedom. Outlet–month panel, 2014–2022. Seventeen outlets span all 108 months and El Heraldo de México, which began publishing in 2017, contributes 72, giving N = 1,908. All columns are Poisson PPML. Cols. (3)–(4) exclude wire copy. Col. (5) is a Poisson PPML of monthly wire articles with log total articles as an offset, and the nine outlet–months with no articles drop. Means cover the period before the transition and are reported in levels.

Relative to right-wing outlets, left-wing outlets employ 31 percent more affiliated journalists after the transition (Column 1), or approximately 39 journalists when estimated in levels. Total articles increase by 50 percent (Column 2), or approximately 850 articles per month. Hard news increases by 51 percent and soft news by 48 percent (Columns 3 and 4). The share of articles from news wires declines by 8 percent, and the estimate is not statistically significant (Column 5). Figure 6 plots the event studies for Columns (1) and (2). The pre-transition coefficients for newsroom size lie between −15 and −2 percent, and none is significant at the five percent level. The pre-transition coefficients for total articles lie between −22 and −16 percent, and one is significant at that level, so the estimate for total articles rests on a weaker pre-transition comparison.

Figure 6: **Event Studies for Newsroom Size and Article Production**



Proportional change $\exp(\hat{\beta}) - 1$ on the interaction of a left-wing outlet indicator with each year relative to the December 2018 transition, for Columns (1) and (2) of Table 5. Event years run from December to November, and the year before the transition is the omitted category, marked by the diamond. Outlet and year-month fixed effects included. Bars are 95 percent confidence intervals from standard errors clustered by outlet across 18 clusters. Dotted lines mark the average coefficient before and after the transition.

Appendix Table F3 decomposes non-wire articles into finer content categories. The largest increases are in crime and security (67 percent), international news (63 percent), and health policy (62 percent). The smallest is in entertainment, where the 19 percent estimate is not statistically significant.

The transition may also have changed private advertising. If private advertisers place advertisements where commercial value is highest, a shift of private advertising toward left-wing outlets would indicate that the transition raised the commercial value of these outlets. Appendix Figure F1 reports the Poisson event study for the number of private advertisements. Relative to right-wing outlets, private advertisements at left-wing outlets increase by 20 percent in the transition year and by 38 percent in the following year. The pooled estimate of 57 percent is not statistically significant. The point estimates suggest that both sources of revenue moved toward left-wing outlets, so the estimates above may reflect both sources and not government advertising alone.

Newsroom size and article counts measure productive capacity and output. Two of these estimates bear on quality. Hard news accounts for approximately 69 percent of

the differential change in non-wire articles, and the share of wire copy does not change significantly. Neither estimate measures how well an outlet reports a given story, so I turn to measures of original reporting.

I measure quality at the level of news events, which separates the number of stories an outlet covers from the detail with which it reports each story. Following Cagé et al. (2020) and Durante et al. (2026), an unsupervised clustering algorithm groups articles by the similarity of their text and their timing, so that each event contains every article in the sample about the same occurrence.

I construct three measures. The first is an indicator for whether the outlet published any article about the event. The second, scoops, indicates whether the outlet's first own article on the event appeared on the event's first day, which measures how early the outlet reaches a story. The third, unique entities, counts the distinct factual elements an outlet contributes to the event, such as a named actor, an organization, a place, or a figure. Appendix F.2 describes the clustering procedure, the extraction of factual elements, and their validation.

Table 6 reports the estimates for all events and separately for hard and soft news events. An event is classified as hard news when at least half of its articles have a hard news dominant topic. Event fixed effects absorb all event-level variation, so the coefficient compares outlets covering the same story.

Relative to misaligned outlets, aligned outlets cover 7.8 percentage points more events after the transition, an increase of 29 percent over the pre-transition mean of 0.27 (Column 1). Unique entities increase by 3.72, or 37 percent of the pre-transition mean of 10.19 (Column 3). Scoops increase by 1.8 percentage points, or 4 percent of the pre-transition mean of 0.49, and the estimate is not statistically significant (Column 2). Among hard news events, coverage increases by 8.1 percentage points, scoops by 2.2 percentage points, and unique entities by 3.62, and the scoops estimate is significant only at the ten percent level. Among soft news events, coverage increases by 7.1 percentage points and unique

entities by 3.99. These estimates are similar in size to those for hard news but are not statistically significant. Aligned outlets therefore covered more events and reported more factual detail, and the estimates are precise only for hard news.

Table 6: **Effect of Left-Wing Government on Journalistic Quality**

	$\mathbf{1}\{\text{Report} > 0\}$ (1)	Scoops (2)	Unique Entities (3)
All Events			
Left-Wing Government × Left-Wing Outlet	0.0775**	0.0177	3.7234**
	(0.0319)	(0.0100)	(1.4349)
Pre-period mean	0.2652	0.4886	10.19
Observations	3,404,704	455,313	2,700,888
Hard News			
Left-Wing Government × Left-Wing Outlet	0.0806**	0.0218*	3.6165**
	(0.0316)	(0.0107)	(1.4567)
Pre-period mean	0.2630	0.5108	10.06
Observations	2,324,382	313,480	1,907,931
Soft News			
Left-Wing Government × Left-Wing Outlet	0.0708	0.0084	3.9903
	(0.0557)	(0.0121)	(3.0899)
Pre-period mean	0.2717	0.4335	10.64
Observations	1,059,863	140,857	779,056
Event FE	✓	✓	✓
Outlet × Topic FE	✓	✓	✓

Note: *p<0.1; **p<0.05; ***p<0.01. Each cell is a separate regression with event and outlet × topic fixed effects, estimated on events covered by at least three outlets, 2014–2022. Pre-period means are for left-wing outlets before December 2018. Col. (2) uses outlet–event pairs with at least one non-wire article, for events lasting more than one day. Col. (3) uses events with at least one extracted factual element. Standard errors clustered by outlet in parentheses. Stars are based on wild cluster bootstrap p-values with the null imposed.

These measures may capture volume rather than quality. The number of events an outlet covers may reflect newsroom capacity rather than newsworthiness, and the number of factual elements may include detail that readers do not value. The measures capture original reporting and the production of new information, which is one dimension of journalistic quality (Turkel et al., 2021; Armona et al., 2024). In addition, Durante et al. (2026) find that the number of facts in an article and its distance from the article that first

reported the story predict which outlet a news aggregator features on its front page, which suggests that readers value these characteristics.

6.2 Circulation

Favoritism alters the information environment only if it affects readers, because an outlet that publishes more without reaching more readers leaves the ideological composition of the market unchanged. I therefore examine circulation at left-wing and right-wing outlets in the balanced panel of circulation by outlet and state. Because circulation is zero in 14 percent of outlet–state cells, I estimate equation 1 by Poisson pseudo-maximum likelihood.

Two channels could increase circulation at left-wing outlets after the transition. On the supply side, the additional revenue financed journalists and articles, and readers may prefer a newspaper with broader coverage. On the demand side, the 2018 election may have shifted readers toward left-wing outlets independently of their content. Only the first channel links circulation to government advertising.

To separate the two channels, I use cross-state variation in the increase in Morena's vote share between 2015 and 2018. Because each newspaper distributes the same edition across states, a content-driven gain should be similar in all states, while a preference-driven gain should be larger where more voters shifted toward Morena. The design relies on two assumptions, that content effects are uniform across states and that demand shifts are proportional to the local electoral shift. I estimate the following triple difference-in-differences specification by Poisson pseudo-maximum likelihood,

$$\begin{aligned} Y_{nst} &= \beta_1 \left(\text{Left-Wing Gov}_t \times \text{Left-Wing Outlet}_n\right) \\ &\quad + \beta_2 \left(\text{Left-Wing Gov}_t \times \text{Left-Wing Outlet}_n \times D_s\right) \\ &\quad + \beta_3 \left(\text{Left-Wing Gov}_t \times D_s\right) + \alpha_{ns} + \gamma_t + \varepsilon_{nst} \end{aligned} \tag{5}$$

where $D_s = \mathbf{1}[\Delta\text{Vote Share Left-Wing Party}_s > \text{median}]$ indicates states with an above-median increase in the left-wing vote share between 2015 and 2018, α_{ns} are outlet-state fixed effects, and γ_t are year fixed effects. Appendix Figure F3 maps the change in Morena's vote share by state. Southern and southeastern states shifted by 50 to 65 percentage points, and northern states by 20 to 35 percentage points. The coefficient β_1 measures the content-driven component common to all states, β_2 measures the additional preference-driven component, and β_3 measures the effect of the electoral shift on circulation regardless of outlet ideology.

Table 7 presents the results. In Column (1), the coefficient on Left-Wing Gov × Left-Wing Outlet is 0.246 (standard error 0.165), which corresponds to 28 percent and is not statistically significant. In Column (2), this coefficient falls to 0.202 (0.164), and the triple interaction with the high-shift indicator is 0.484 (0.230), with a p-value of 0.051. The coefficient on Left-Wing Gov × $\mathbf{1}\{\Delta$ Left-Wing Vote Share > median$\}$ is −0.071, so the electoral shift does not increase circulation across all outlets. Column (3) replaces the indicator with the continuous electoral shift in standard deviations, and the triple interaction is 0.221 (0.131). The circulation gain is therefore concentrated in the states where Morena gained most, which is more consistent with a preference-driven component than with a content-driven gain common to all states.

Table 7: **Decomposing Circulation Gains**

	Circulation (Poisson)		
	(1)	(2)	(3)
Left-Wing Gov × Left-Wing Outlet	0.246	0.202	0.439**
	(0.165)	(0.164)	(0.189)
Left-Wing Gov × Left-Wing Outlet × **1**{Δ Left-Wing Vote Share > median}		0.484*	
		(0.230)	
Left-Wing Gov × **1**{Δ Left-Wing Vote Share > median}		−0.071	
		(0.128)	
Left-Wing Gov × Left-Wing Outlet × Δ Left-Wing Vote Share (SD)			0.221
			(0.131)
Left-Wing Gov × Δ Left-Wing Vote Share (SD)			−0.022
			(0.061)
$\exp(\hat{\beta}) - 1$, Left-Wing Gov × Left-Wing Outlet	28%	22%	55%
Outlet × State FE	✓	✓	✓
Year FE	✓	✓	✓
Observations	4,528	4,528	4,528

Note: *p<0.1; **p<0.05; ***p<0.01. Poisson PPML of outlet–state circulation, 2014–2022. Standard errors are clustered on outlet, with p-values from a t distribution with 17 degrees of freedom. Outlet–state cells with zero circulation in every year drop. **1**{Δ Left-Wing Vote Share > median} marks states whose increase in left-wing vote share between 2015 and 2018 exceeded the median.

The circulation estimates do not establish whether the increase in production reached readers. The average gain is imprecisely estimated and concentrates in the states where Morena gained most, which is consistent with a shift in demand after the election. Web visits increase by 138 percent at left-wing outlets relative to right-wing outlets (Appendix Figure F2). The evidence establishes the step before circulation. Ministries directed advertising toward aligned outlets, and these outlets increased their production relative to misaligned outlets.

7 Conclusion

Existing accounts of media capture emphasize government interference with news content, through censorship, bribery, and quid pro quo arrangements. In consolidated democracies, constitutional guarantees of press freedom and market discipline constrain such interference, since readers can sanction outlets that provide biased coverage by switching

to competitors. This paper examines an alternative mechanism of influence that operates through the market, namely political favoritism in the allocation of government advertising. Rather than altering the content of news, the government creates an asymmetry in resources across competing outlets that affects their capacity to expand or contract.

Using data from Mexico, the paper documents how a government deploys this mechanism. Following the 2018 transition, ideologically aligned outlets became 18.3 percentage points more likely to receive a contract from a given ministry. Aligned outlets received $20,876 more per campaign than the allocation that maximizes the reach of each campaign's stated demographic, approximately 40 percent of their average allocation, and the deviation relative to an allocation that targets the incumbent's voters is of similar magnitude. The reallocation therefore reflects ideological favoritism rather than the pursuit of particular audiences, and it entailed an informational cost, since campaigns informed between 23 and 36 percent of the target readers attainable with the same budgets. Following the transition, newsroom size at left-wing outlets increased by 31 percent relative to right-wing outlets, and the number of articles by 50 percent. Left-wing outlets also increased their coverage of news events and the factual detail of their reporting relative to right-wing outlets.

Favoritism in the allocation of government advertising operates within democratic institutions. It alters the relative capacity of competing outlets to produce news whether or not their editorial content changes. Whether favoritism yields electoral returns for the incumbent remains an open question. The features of a pluralistic market that constrain censorship and bribery do not constrain this channel, because it operates through competition among outlets rather than through the content of any one of them.

References

Adena, Maja, Ruben Enikolopov, Maria Petrova, Veronica Santarosa, and Ekaterina

Zhuravskaya, "Radio and the Rise of the Nazis in Prewar Germany," *Quarterly Journal of Economics*, 2015, *130* (4), 1885–1939.

Angelucci, Charles and Julia Cagé, "Newspapers in times of low advertising revenues," *American Economic Journal: Microeconomics*, 2019, *11* (3), 319–364.

Armona, Luis, Matthew Gentzkow, Emir Kamenica, and Jesse M. Shapiro, "What is Newsworthy? Theory and Evidence," *American Economic Review: Insights*, 2024. Forthcoming.

Berry, Christopher R., Barry C. Burden, and William G. Howell, "The President and the Distribution of Federal Spending," *American Political Science Review*, 2010, *104* (4), 783–799.

Besley, Timothy and Andrea Prat, "Handcuffs for the grabbing hand? Media capture and government accountability," *American Economic Review*, 2006, *96* (3), 720–736.

Bor, Alexander, Gabor Simonovits, and Milan W. Svolik, "Choice versus Control: Government Capture of the Media and the Consumption of Political News," 2026. Working paper.

Brollo, Fernanda and Tommaso Nannicini, "Tying Your Enemy's Hands in Close Races: The Politics of Federal Transfers in Brazil," *American Political Science Review*, 2012, *106* (4), 742–761.

Burgess, Robin, Remi Jedwab, Edward Miguel, Ameet Morjaria, and Gerard Padró i Miquel, "The value of democracy: Evidence from road building in Kenya," *American Economic Review*, 2015, *105* (6), 1817–1851.

Cagé, Julia, "Media competition, information provision and political participation: Evidence from French local newspapers and elections, 1944–2014," *Journal of Public Economics*, 2020, *185*, 104077.

_ , **Nicolas Hervé, and Marie-Luce Viaud**, "The Production of Information in an Online World," *The Review of Economic Studies*, 2020, *87* (5), 2126–2164.

Chen, Jiafeng and Jonathan Roth, "Logs with Zeros? Some Problems and Solutions," *Quarterly Journal of Economics*, 2024, *139* (2), 891–936.

Durante, Ruben and Brian Knight, "Partisan control, media bias, and viewer responses: Evidence from Berlusconi's Italy," *Journal of the European Economic Association*, 2012, *10* (3), 451–481.

_ , **Ruben Enikolopov, Daniil Mikhailov, and Andrey Simonov**, "Platform Power in News Aggregation," 2026. Working paper.

Enikolopov, Ruben, Maria Petrova, and Ekaterina Zhuravskaya, "Media and political persuasion: Evidence from Russia," *American Economic Review*, 2011, *101* (7), 3253–3285.

Gehlbach, Scott and Konstantin Sonin, "Government control of the media," *Journal of public Economics*, 2014, *118*, 163–171.

Gentzkow, Matthew and Jesse M Shapiro, "What drives media slant? Evidence from US daily newspapers," *Econometrica*, 2010, *78* (1), 35–71.

_ , _ , **and Matt Taddy**, "Measuring group differences in high-dimensional choices: method and application to congressional speech," *Econometrica*, 2019, *87* (4), 1307–1340.

_ , **Jesse M. Shapiro, Frank Yang, and Ali Yurukoglu**, "Pricing Power in Advertising Markets: Theory and Evidence," *American Economic Review*, 2024, *114* (2), 500–533.

Grossman, Gene M. and Carl Shapiro, "Informative Advertising with Differentiated Products," *The Review of Economic Studies*, 1984, *51* (1), 63–81.

Hodler, Roland and Paul A. Raschky, "Regional Favoritism," *The Quarterly Journal of Economics*, 2014, *129* (2), 995–1033.

King, Gary, Jennifer Pan, and Margaret E. Roberts, "How Censorship in China Allows Government Criticism but Silences Collective Expression," *American Political Science Review*, 2013, *107* (2), 326–343.

Knight, Brian and Ana Tribin, "The limits of propaganda: Evidence from Chavez's Venezuela," *Journal of the European Economic Association*, 2019, *17* (2), 567–605.

Kronick, Dorothy and John Marshall, "Collateral Censorship: Theory and Evidence from Venezuela," 2024. Working paper.

Lawson, Chappell, *Building the fourth estate: Democratization and the rise of a free press in Mexico*, Univ of California Press, 2002.

Lee, David S., "Training, Wages, and Sample Selection: Estimating Sharp Bounds on Treatment Effects," *Review of Economic Studies*, 2009, *76* (3), 1071–1102.

McMillan, John and Pablo Zoido, "How to subvert democracy: Montesinos in Peru," *Journal of Economic perspectives*, 2004, *18* (4), 69–92.

Padró i Miquel, Gerard, "The Control of Politicians in Divided Societies: The Politics of Fear," *The Review of Economic Studies*, 2007, *74* (4), 1259–1274.

Paxman, Andrew, *Mexican Watchdogs: The Rise of a Critical Press since the 1980s*, Chapel Hill: University of North Carolina Press, 2025.

Roberts, Margaret E., Brandon M. Stewart, and Dustin Tingley, "stm: An R Package for Structural Topic Models," *Journal of Statistical Software*, 2019, *91* (2), 1–40.

_ , _ , _ , Christopher Lucas, Jetson Leder-Luis, Shana Kushner Gadarian, Bethany Albertson, and David G. Rand, "Structural Topic Models for Open-Ended Survey Responses," *American Journal of Political Science*, 2014, *58* (4), 1064–1082.

Rosenstiel, Leah, "The Distributive Politics of Grants-in-Aid," *American Political Science Review*, 2025, *119* (1), 152–164.

Snyder Jr., James M and David Strömberg, "Press coverage and political accountability," *Journal of Political Economy*, 2010, *118* (2), 355–408.

Strömberg, David, "Radio's impact on public spending," *Quarterly Journal of Economics*, 2004, *119* (1), 189–221.

Szeidl, Adam and Ferenc Szucs, "Media capture through favor exchange," *Econometrica*, 2021, *89* (1), 281–310.

Tella, Rafael Di and Ignacio Franceschelli, "Government advertising and media coverage of corruption scandals," *American Economic Journal: Applied Economics*, 2011, *3* (4), 119–151.

Turkel, Eray, Anish Saha, Rhett Carson Owen, Gregory J Martin, and Shoshana Vasserman, "A method for measuring investigative journalism in local newspapers," *Proceedings of the National Academy of Sciences*, 2021, *118* (30), e2105155118.

A Setting

A.1 Regulatory Framework for Government Advertising

Two obligations govern federal advertising over the sample period, and both were in force throughout it. A 2007 reform to Article 134 of the Constitution requires that official advertising be institutional in character and bars it from carrying the name, image, voice, or symbol of any public servant. From 2003 onward, an annual *Acuerdo* issued each December by the Secretaría de Gobernación required every federal entity to file an annual communication program, obtain prior authorization for each campaign, exhaust official airtime before buying commercial space, register the outlets it contracts with, and report what it spent.

The *Ley General de Comunicación Social*, published on 11 May 2018 and in force from 1 January 2019, gave that apparatus statutory standing. It extended the rules to state and municipal government (Art. 3), fixed the annual planning cycle (Arts. 22–25), placed the media registry in law (Arts. 35–37), and set the disclosure requirements (Arts. 40–43). Two 2022 reforms, in April and December, changed language and state-level spending limits without touching the allocation rules. The substantive obligations a ministry faces therefore do not change at the 2018 transition, which matters for the design in Section 4: the legal environment is constant across the comparison, so nothing in the reallocation I document can be attributed to a change in what the law permitted.

Article 8 of the law requires that ministries select outlets by cost efficiency, defined as the cost of reaching the campaign's stated audience. This is the legal standard the cost-efficiency benchmark in Section 5 operationalizes. The law names no auditor for that requirement and attaches no penalty to departing from it, which is why the deviations I measure are best read as discretion rather than error.

A.2 Government Advertising Descriptives

The federal government is a large source of advertising for these outlets. Figure A3 plots the share of each outlet's advertising that comes from the federal government, and the median outlet-year stands at 25 percent.

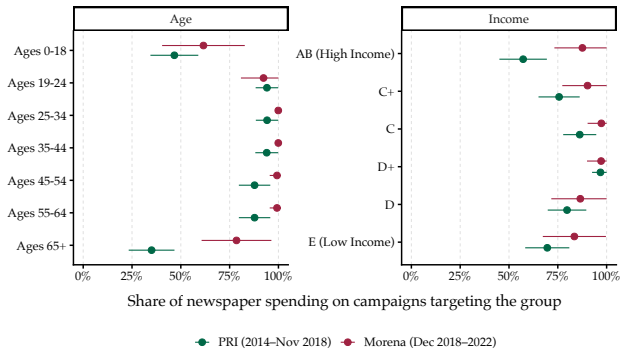
Figure A4 plots average government advertising spending per outlet in each semester, by outlet ideology. Spending falls for both groups of outlets after the December 2018 transition. It also peaks in the second semester of every year, because Congress appropriates advertising budgets annually and ministries forfeit whatever they have not committed by December, so they place the unspent balance before the fiscal year closes.

Table A1: **Government Advertising Budgets by Ministry and Party (Millions MXN)**

Ministry	PRI		Morena	
	Millions	%	Millions	%
Health	361.82	21.0	366.14	46.0
Finance	235.09	14.0	90.99	11.0
Education	228.66	14.0	1.36	0.0
Energy	121.22	7.0	91.60	11.0
Social Development	180.11	11.0	29.30	4.0
Environment	141.00	8.0	48.56	6.0
Communications and Transportation	104.98	6.0	21.91	3.0
Labor	83.58	5.0	15.26	2.0
Interior	35.96	2.0	31.21	4.0
Navy	34.32	2.0	30.68	4.0
Agrarian, Territorial and Urban Development	59.37	4.0	0.00	0.0
National Defense	10.85	1.0	37.36	5.0
Foreign Affairs	29.87	2.0	5.08	1.0
Presidency	20.41	1.0	8.07	1.0
Culture	12.28	1.0	11.22	1.0
Economy	15.18	1.0	0.00	0.0
Agriculture	7.50	0.0	6.16	1.0
Tourism	8.65	1.0	1.15	0.0
Public Administration	0.00	0.0	1.07	0.0

The table shows government advertising expenditures during the two administrations across federal ministries. Total spending is reported in millions of Mexican pesos, and the percentage represents the proportion of each ministry's budget for a given administration.

Figure A1: **Share of Advertising Spending on Campaigns Targeting Each Group, by Government**



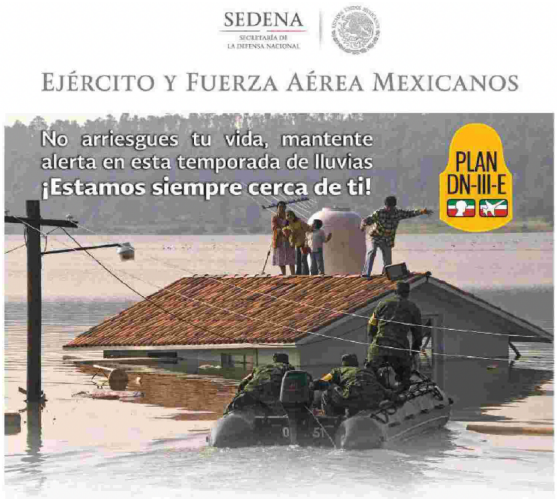
Each dot is the share of federal advertising spending in the 18 sample newspapers that goes to campaigns whose stated target population includes the group, by government. A campaign can target several groups, so shares do not sum to 100 percent. Income groups are the socioeconomic levels reported in each campaign's annual communication program. Bars are 95 percent confidence intervals.

Table A2: **Government Advertising Campaigns Run Under Both Governments**

Ministry	Type	Title	Total Spending (USD)	
			PRI	Morena
Health	Prevention	Obesity Prevention	1,946,087	2,786,190
Interior	Civic Culture	Human Rights	66,200	1,895,410
Navy	Government Services	National Maritime Authority	1,325,999	157,456
Navy	Civic Culture	The Navy Close to You	1,181,894	131,892
Defense	Civic Culture	Military Commemorations	891,358	276,259
Foreign Affairs	Government Services	Protection of Mexicans Abroad	1,089,583	35,021
Education	Civic Culture	Sports in Mexico	1,087,621	25,974
Defense	Civic Culture	Commitment to Mexico	432,652	379,743
Defense	Civic Culture	Army and Family Outreach	384,237	58,197
Presidency	Civic Culture	In Search of Equality	153,061	209,533
Tourism	Prevention	Green Angels Road Service	302,958	11,356
Culture	Civic Culture	Linguistic Diversity	164,744	145,441
Defense	Civic Culture	Military Education	214,854	44,258
Tourism	Transparency	Local Tourism Promotion	214,352	23,920
Energy	Civic Culture	Daylight Saving Time	109,721	4,525
Education	Civic Culture	National Sports Award	2,710	18,701

The table shows government advertising campaigns that did not change their target audience during the sample period. Spending during the PRI administration covers 2014 to November 30, 2018; spending during the Morena administration covers December 1, 2018 to 2022.

Figure A2: **Example of a Cross-Administration Campaign at the Ministry of Defense**

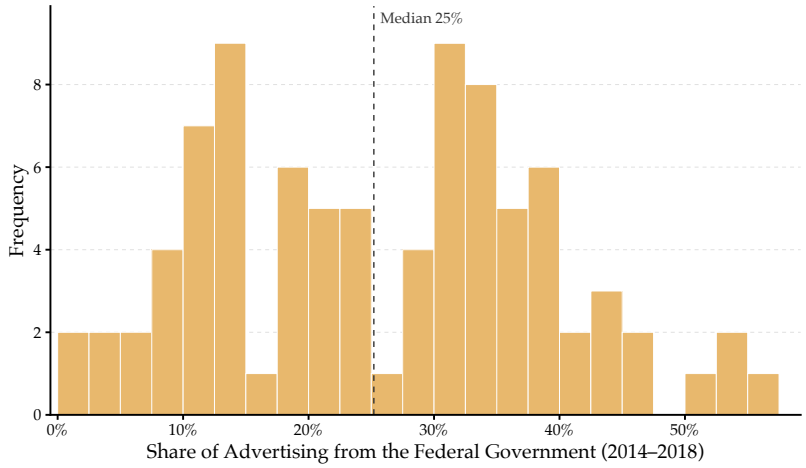


(a) **PRI Administration (2014–2018)**



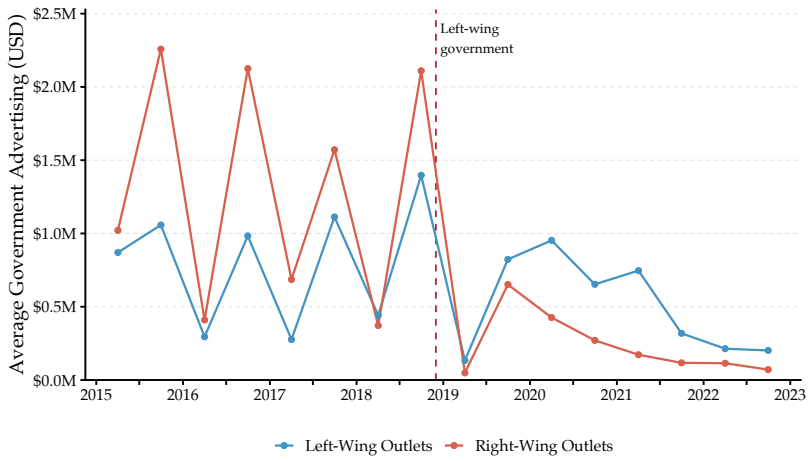
(b) **Morena Administration (2018–2022)**

Figure A3: **Share of Advertising from the Federal Government**



An observation is an outlet-year, 2014–2018. Includes only federal government advertisers, excluding other public advertisers.

Figure A4: **Government Advertising Spending by Outlet Ideology**



Average semester spending on ministry advertising campaigns per outlet, by outlet ideology (9 left-wing, 9 right-wing). Spending is deflated to 2022 pesos and converted to USD at the 2022 average exchange rate. The dashed line marks the December 2018 presidential transition from PRI to Morena.

B Data

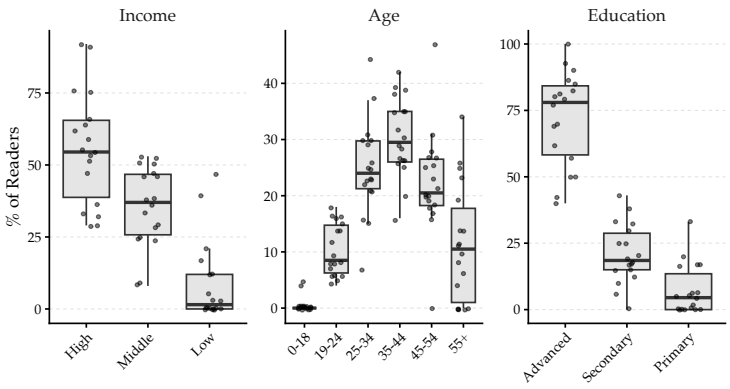
B.1 Media Market

Table B1: **Summary Statistics for News Articles**

Newspaper	N Total Articles	Articles per Year	
		Mean	SD
La Prensa	198,133	22,014.78	5,148.32
Esto	129,351	14,372.33	8,254.49
Ovaciones	245,451	27,272.33	7,839.42
Reporte Indigo	81,175	9,019.44	1,916.97
Publimetro	72,942	8,104.67	2,795.21
El Universal	399,328	44,369.78	7,629.34
Reforma	622,705	69,189.44	9,873.62
La Jornada	397,990	44,221.11	3,524.11
Excelsior	545,266	60,585.11	9,675.98
Milenio	339,008	37,667.56	8,352.47
La Razon	59,330	17,703.33	2,460.09
El Financiero	203,748	22,638.67	5,182.94
24 Horas	157,708	17,523.11	2,282.09
La Crónica de Hoy	253,724	28,191.56	8,771.31
El Sol de México	373,994	41,554.89	13,712.01
El Heraldo	143,123	17,890.38	11,760.17
Diario de México	83,352	9,261.33	3,337.80
El Economista	261,679	29,075.44	3,260.44

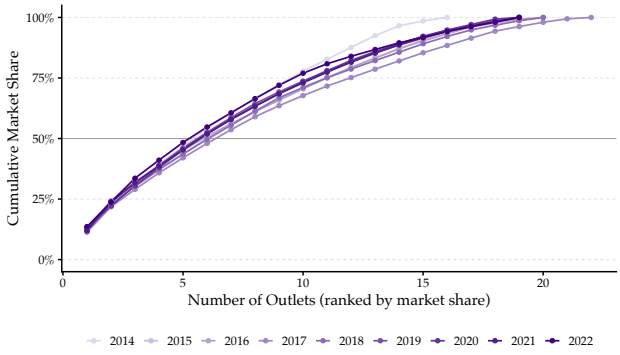
This table reports summary statistics of news articles for 18 national newspapers from 2014 to 2022. N Total Articles shows the total number of articles published by each outlet over the entire period. Articles per Year Mean and SD report the average number of articles per year and the corresponding standard deviation.

Figure B1: **Reader Demographics Distribution Across Newspaper Outlets (2022)**



This figure displays the distribution of reader demographics across 18 newspaper outlets in 2022. Each panel shows boxplots representing the variation in the share of readers within each demographic category. Income levels are classified as High (AB/C+), Middle (C/D+), and Low (D/E). Education categories include Advanced, Secondary, and Primary education.

Figure B2: **Cumulative Market Share (2014–2022)**



This figure shows the cumulative distribution of market share across the universe of national print media outlets for each year from 2014 to 2022. Outlets are ranked within each year by their circulation, from largest to smallest.

Table B2: **Ownership Structure of National Newspapers**

Newspaper	In sample	Owner	Conglomerate	**Market share**	
				Mean	SD
El Grafico		Ealy Ortíz	Grupo Universal	12.30	0.75
La Prensa	1	Vázquez Raña	OEM	11.50	1.13
Esto	1	Vázquez Raña	OEM	7.57	0.80
Ovaciones	1	Vázquez Raña	OEM	6.81	0.75
Reporte Indigo	1	Garza García	Mac Multimedia	6.69	1.29
Publimetro	1	Rivas Garza	Publicaciones Metropolitanas	6.66	0.83
El Universal	1	Ealy Ortíz	Grupo Universal	6.16	1.59
Reforma	1	Junco de la Vega	Grupo Reforma	6.09	0.66
La Jornada	1	–	Demos	5.23	1.81
Mas por Mas		–	–	4.85	2.52
Uno Mas Uno		Libien Kaui	–	4.77	1.35
Excelsior	1	Vázquez Aldir	Grupo Imagen Multimedia	4.34	1.03
Milenio	1	González Family	Multimedios	4.32	0.47
La Razon	1	Garza Cantú	LRHG Informativo	4.19	0.65
El Financiero	1	Arroyo Rodríguez	Grupo Lauman	3.83	0.89
24 Horas	1	Zabludovsky	Información Integral 24/7	3.67	1.13
La Crónica de Hoy	1	Kahwagi Gastine	La Crónica	3.16	0.50
El Sol de México	1	Vázquez Raña	OEM	2.91	0.63
El Heraldo	1	Mieres Zimmermann	Grupo Andrade	2.09	0.18
Diario de México	1	–	Editorial DDM	1.85	0.21
El Economista	1	Nacer Gobera	–	1.73	0.37
Eje Central		Oliver Fernández	OFEM Media Group	0.63	0.09

Market share is computed as the outlet's share of total national print circulation in a given year, averaged over 2014–2022. Ownership and conglomerate information is obtained from the National Print Media Registry (PNMI) and the Media Ownership Monitor Mexico database. When ownership information is not reported, this reflects either the absence of a clearly identifiable controlling shareholder or a cooperative ownership structure.

Table B3: **Classification of Journalists (2014–2022)**

Employment Pattern	**Category**	**N**	**%**
Exclusive Affiliation	*Total*	2,560	**43.10**
	Left-Wing Outlets	1,163	12.97
	Right-Wing Outlets	1,397	15.58
Primary Affiliation	*Total*	2,214	**37.27**
	Left-Wing Outlets	869	9.69
	Right-Wing Outlets	1,345	15.00
Outlet Movers	*Total*	380	**6.39**
	Within-Ideology Moves	214	2.38
	Left → Right	78	0.87
	Right → Left	75	0.84
	Multiple Switches	14	0.16
Freelance/Multi-Outlet		785	**13.21**
Total		**5,939**	**100.00**

This table summarizes the classification of 5,939 journalists in the database. *Exclusive Affiliation* journalists publish exclusively at one outlet. *Primary Affiliation* journalists publish predominantly at one outlet but occasionally contribute to others. *Outlet Movers* switched between outlets during the period, either within the same ideological category or across ideologies. *Freelance/Multi-Outlet* exhibit diverse publication patterns across multiple outlets.

B.2 Constructing the Market Advertising Price

The market advertising price p_{it} is the price a private buyer pays outlet i for one square centimetre of space in year t. It enters the benchmark of Section 5 as the cost at which a ministry converts spending into insertions, so it has to be observed independently of what the government paid.

I take it from the *tarifarios* published quarterly by mediavyasa, a commercial media directory that reproduces the published price list of each outlet. The source covers editions 86 to 121, one per quarter from 2014Q1 to 2022Q4 with no gaps. From each edition I extract the Mexico City daily-newspaper block of the *Prensa* section, which is where the national titles appear.[12]

Each quoted price is a full-page rate. I divide it by the outlet's page area to obtain a price per square centimetre, deflate to 2022 pesos, and convert at the 2022 average exchange rate of 20.11 pesos per dollar. Averaging within outlet and year gives a panel of 157 outlet-years across the 18 outlets in the sample. Coverage is not complete: an outlet is missing from a quarter when it does not appear in that edition's CDMX daily block, and I audit every missing cell against the source PDF rather than imputing it.

Two features of the series matter for how it is used. First, it is a published list price, quoted before any negotiated discount, so it is an upper bound on what a private buyer transacts at. Because the benchmark converts a ministry's budget into insertions by dividing by this price, an upper bound on the price is a lower bound on the insertions a budget could have bought, and therefore a conservative estimate of the reach the observed allocation forgoes. Second, it is quoted for a generic full page. Appendix E.1 reports what position and colour add to that price and shows the range is too narrow to account for the gap between what ministries paid and the market rate.

[12] The block is headed *México, D.F.* in editions 86–97, *CDMX* in 98–108, and *Ciudad de México* in 109–121. Blocks covering the state press and the CDMX weeklies use a different column set and are not extracted. The edition number maps onto the calendar quarter as year $= 2014 + \lfloor (e-86)/4 \rfloor$ and quarter $= (e-86) \bmod 4 + 1$, verified against the PDF creation dates at editions 86, 100, 102 and 121.

Averaged over the sample the market advertising price is \$7.83 per square centimetre under the PRI and \$7.52 under Morena, against \$21.93 and \$13.41 paid by the government, as Table 1 reports.

C Measurement of Ideology

C.1 Alternative Ideology Measures

Table C1 replicates the main analysis using the continuous ideology score rather than the binary classification. Columns (1) and (2) use all campaigns and Columns (3) and (4) use campaigns run under both governments, with the any-contract and spending outcomes of Table 3. The interaction term is Left-Wing Gov × Outlet's Ideology Score, where higher scores indicate more right-leaning positions. A negative coefficient indicates that more right-wing outlets receive less advertising under the left-wing government, consistent with ideological favoritism.

Table C1: **Robustness to a Continuous Ideology Score**

	All Campaigns		**Campaigns Run Under Both Governments**	
	Any contract (1)	**Spending** (2)	**Any contract** (3)	**Spending** (4)
Left-Wing Gov × Outlet's Ideology Score	−0.193***	−160,863***	−0.095**	−13,765**
	(0.067)	(41,466)	(0.038)	(4,834)
Mean (Pre-2018, Left)	0.56	107,562	0.19	4,449
Mean (Pre-2018, Right)	0.78	160,559	0.29	9,696
Year FE	✓	✓	✓	✓
Outlet × Ministry FE	✓	✓	✓	✓
Campaign FE			✓	✓
Observations	1,410	1,410	2,397	2,397

Cols. (1)–(2) restrict to the 10 ministries with campaigns run under both governments. Cols. (3)–(4) use the 17 campaigns run under both governments. Outlet's Ideology Score is the continuous score; higher values indicate more right-leaning positions. SEs clustered at the newspaper level. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$.

Table C2 replicates the analysis using a three-category ideology classification. Outlets are divided into terciles of the continuous score (6 outlets each), namely left-wing, centrist, and right-wing, and right-wing outlets are the reference category. Left-wing outlets gain the most in every column. The centrist coefficients are smaller, and only the coefficient for any contract across all campaigns separates from zero.

Table C2: **Robustness to a Three-Category Ideology Classification**

	All Campaigns		**Campaigns Run Under Both Governments**	
	Any contract	**Spending**	**Any contract**	**Spending**
	(1)	(2)	(3)	(4)
Left-Wing Gov × Left-Wing Outlet	0.342***	169,423**	0.150***	11,565*
	(0.082)	(67,662)	(0.051)	(6,584)
Left-Wing Gov × Centrist Outlet	0.208***	81,700	0.082	−464
	(0.047)	(56,405)	(0.048)	(3,948)
Mean (Pre-2018, Left)	0.56	107,562	0.19	4,449
Mean (Pre-2018, Right)	0.78	160,559	0.29	9,696
Year FE	✓	✓	✓	✓
Outlet × Ministry FE	✓	✓	✓	✓
Campaign FE			✓	✓
Observations	1,410	1,410	2,397	2,397

Reference category: right-wing outlets (6 most right-wing by score). Left, centrist, and right-wing categories are terciles of 18 outlets ranked by the continuous score (6 outlets each). Cols. (1)–(2) restrict to the 10 ministries with campaigns run under both governments. Cols. (3)–(4) use the 17 campaigns run under both governments. SEs clustered at the newspaper level. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$.

C.1.1 Outlets Scored Without Editorials

Diario de México, El Heraldo de México, and Esto publish no editorial section and are scored by applying the estimated phrase loadings to their news articles, as described in Section 3.2. Two features of these corpora differ from the editorial corpora used for the other fifteen outlets. They are far larger, with 26,493 articles for Esto against a median of 469 editorials per outlet, and they contain agency copy, which carries no editorial

stance. Table C3 addresses both. I first restrict each corpus to journalist-authored content, dropping wire and agency material, and then draw 500 bootstrap samples at the median and mean editorial corpus size, rescoring each draw.

The journalist restriction moves each score by less than 0.02, and the bootstrap confirms that corpus size is not what drives the classification of Diario de México or Esto, whose signs are stable in 99.8 and 100 percent of draws respectively. El Heraldo de México is genuinely ambiguous. Its score sits within 0.03 of the anchor midpoint and its sign is stable in only 78 to 83 percent of draws. It is coded as left-wing in the main analysis. The outlet enters a single ministry–outlet series and the main estimates are unchanged when it is coded as right-wing.

Table C3: **Bootstrap Validation for Outlets Scored From News Articles**

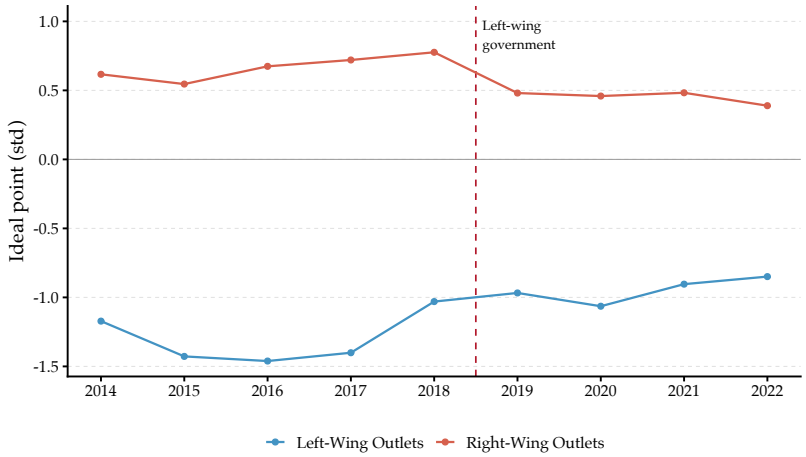
	Articles (all)	**Journalist articles**	**Score (all)**	**Score (journalist)**	**95% CI (bootstrap)**	**Sign stable**
Diario de México	5,102	4,539	+0.114	+0.102	[+0.025, +0.174]	99.8%
El Heraldo de México	3,081	2,903	−0.034	−0.027	[−0.097, +0.051]	77.8%
Esto	26,493	21,502	−0.296	−0.296	[−0.347, −0.245]	100.0%

Scores are recentered at the anchor midpoint, so negative values indicate left-leaning language. Journalist articles exclude wire and agency copy. Bootstrap draws 500 samples of 469 articles, the median editorial corpus size, from the journalist-authored corpus and rescores each draw. Sign stable reports the share of draws whose sign matches the full-corpus classification. At the mean editorial corpus size of 596 articles the corresponding shares are 99.8, 82.6, and 100.0 percent.

C.2 Stability of the Ideology Scores over Time

Figure C1 applies the phrase loadings estimated on the 2014–2017 editorials to the editorials published in each year from 2014 to 2022, for the fifteen outlets with editorial sections.

Figure C1: **Editorial Slant by Outlet Ideology, 2014–2022**



The figure applies the loadings of Figure 1 to editorials published in each year and plots the article-weighted mean score of left-wing and right-wing outlets.

C.3 Ideology and Audience Composition

I compare the ideology score against the audience composition each outlet reports to the National Print Media Registry. The profiles cover three income brackets, six age brackets, three education levels, sex, and an urban–rural split, and are available for all 18 sample outlets in every year from 2014 to 2022. Figure C2 presents the comparison.

Audience similarity is unrelated to ideological similarity. Across the 153 outlet pairs the correlation between audience distance and the gap in ideology scores is −0.025, with a Mantel permutation p-value of 0.87. Pairs on the same side of the ideological divide are no closer in audience space than pairs on opposite sides, with mean distances of 5.46 and 5.53 respectively.

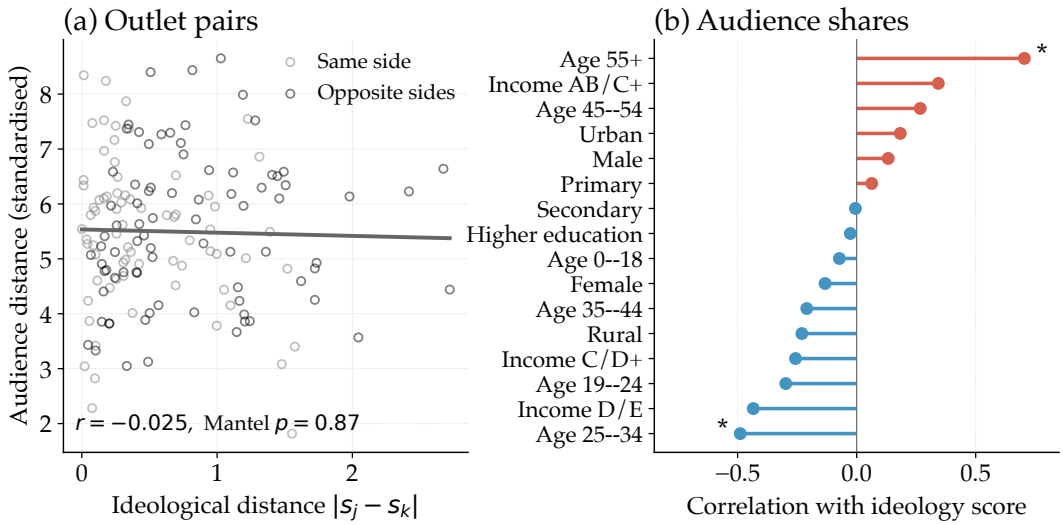
Among individual dimensions, only age is strongly related to the score. Right-leaning outlets draw older readers, with a correlation of +0.71 for the share aged 55 and above and −0.49 for the share aged 25 to 34. Income runs in the expected direction but weakly, at +0.34 for the highest bracket and −0.43 for the lowest, neither significant in a cross-section of 18 outlets. Education correlations are below 0.03 in absolute value.

The audience space is organized by outlet genre rather than politics. The first two

principal components account for 28 and 23 percent of the variance, and the first is uncorrelated with ideology at −0.01. It separates the tabloid and sports titles *La Prensa* and *Esto* from the business and broadsheet titles *El Financiero* and *Excélsior*, while *La Jornada* and *Reforma*, the two ideological poles of the sample, sit close together. The second component correlates with the score at +0.34, reflecting the age gradient.

The measure is therefore not a proxy for readership, and the near-independence of alignment and audience composition supports the identification argument in Section 4. A government reallocating advertising toward demographic groups it wishes to reach would not, as a byproduct, reallocate toward ideologically aligned outlets, because the two dimensions do not line up. Age is the exception, and remains a live alternative explanation for any result that operates through the age composition of readers.

Figure C2: **Ideological Similarity and Audience Composition**



Audience shares reported to the National Print Media Registry, averaged over 2014–2022. Panel (a) plots audience distance against ideological distance for the 153 outlet pairs, with a linear fit and a Mantel permutation p-value. Panel (b) correlates each audience share with the ideology score; asterisks mark significance at 5 percent. Higher scores are more right-leaning.

D Robustness for the Allocation Estimates

D.1 Alternative Estimators for the Intensive Margin

Following Chen and Roth (2024), Table D1 reports two alternative estimators for the intensive margin. Column (1) presents a Poisson PPML regression, which is scale-invariant, and $\exp(\hat{\beta}) - 1 \approx 222$ percent represents the proportional increase in expected spending for aligned outlets. Column (2) reports Lee bounds (Lee, 2009) on the effect for ministry–outlet pairs that receive positive advertising under both administrations. The bounds range from −$91,886 to $188,717 and contain zero.

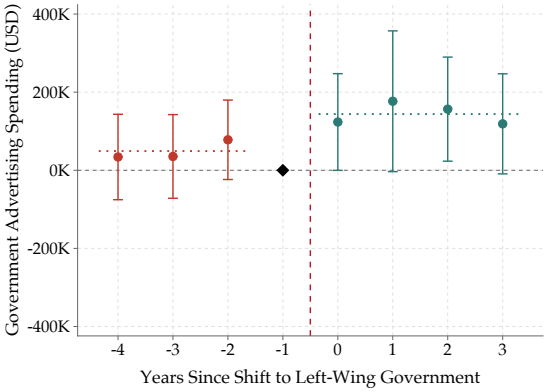
Table D1: **Intensive Margin Robustness Under Poisson and Lee Bounds**

	Poisson PPML (semi-elasticity) (1)	**Lee Bounds** (2022 USD, conditional intensive margin) (2)
Left-Wing Gov × Left-Wing Outlet	1.169**	[−91,886 ; +188,717]
	(0.452)	
Year FE	✓	
Outlet × Ministry FE	✓	
Trimming share		0.444
Observations	1,149	771

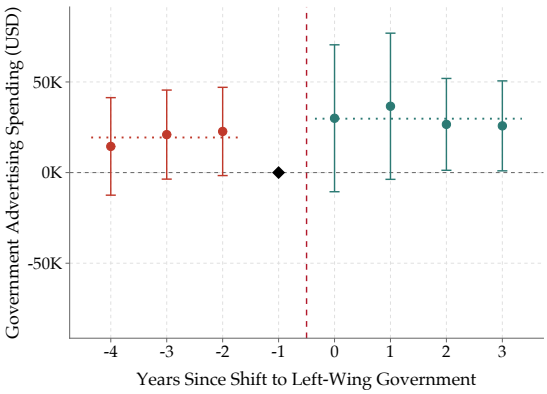
Sample restricted to the 10 ministries with campaigns run under both governments. Poisson drops outlet–ministry cells with all-zero outcomes; $\exp(1.169) - 1 \approx 222\%$. *Lee bounds identify the average treatment effect on the conditional intensive margin under a monotonicity assumption (Lee, 2009). SE clustered at the newspaper level.* $^{**}p<0.05$, $^{*}p<0.1$.

D.2 Event Studies

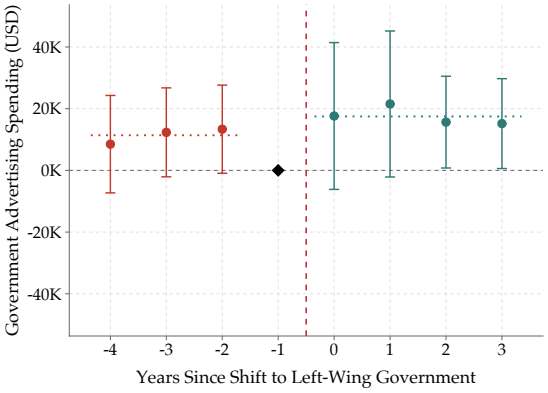
Figure D1: **Event Studies of Differential Government Advertising by Outcome**



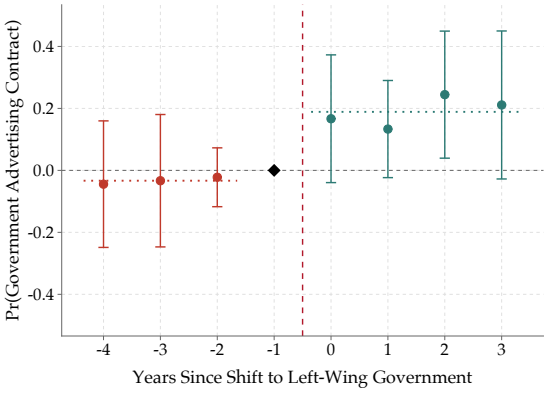
(a) **Outlet–Ministry, Spending (USD)**



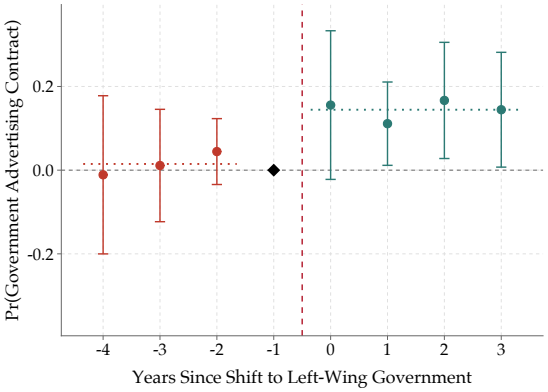
(b) **Outlet–Ministry, Campaigns Run Under Both Governments, Spending (USD)**



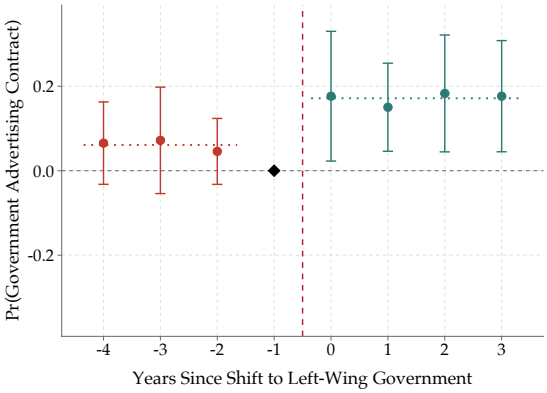
(c) **Campaign–Outlet, Spending (USD)**



(d) **Outlet–Ministry, Any Spending**



(e) **Outlet–Ministry, Campaigns Run Under Both Governments, Any Spending**



(f) **Campaign–Outlet, Any Spending**

This figure presents event study estimates for the any-contract and spending outcomes. All panels include outlet–ministry or outlet–campaign fixed effects. Coefficients represent the interaction between event time (years relative to the December 2018 government transition) and outlet ideology. Error bars show 95% confidence intervals with standard errors clustered at the outlet level.

D.3 Fixed Effects, Inference, and Outlet Controls

Table D2 re-estimates the two outcomes of Table 3 under ministry-by-year fixed effects and alternative inference. Table D3 adds pre-2018 outlet characteristics interacted with year fixed effects. Across those specifications the coefficient on any contract ranges from 0.174 to 0.241 and the coefficient on spending ranges from $80,939 to $118,576.

Table D2: **Robustness to Fixed Effects and Inference**

	Baseline	Ministry × Year FE	Randomization inference	Two-way clustering	Outlet × Year clustering
	(1)	(2)	(3)	(4)	(5)
Panel A. Any contract					
Left-Wing Government × Left-Wing Outlet	0.183**	0.183**	0.183**	0.183**	0.183***
	(0.082)	(0.084)	(0.082)	(0.077)	(0.047)
Cluster-robust p	[0.040]	[0.044]	[0.040]	[0.049]	[0.000]
Randomization inference p		[0.049]	[0.049]		
Wild cluster bootstrap p	[0.041]	[0.041]			
Panel B. Spending (2022 USD)					
Left-Wing Government × Left-Wing Outlet	105,362*	105,362*	105,362*	105,362*	105,362***
	(53,749)	(55,186)	(53,749)	(50,655)	(30,437)
Cluster-robust p	[0.067]	[0.073]	[0.067]	[0.076]	[0.001]
Randomization inference p		[0.052]	[0.052]		
Wild cluster bootstrap p	[0.044]	[0.044]			
Year FE	✓		✓	✓	✓
Ministry × Year FE		✓			
Outlet × Ministry FE	✓	✓	✓	✓	✓
Clustering	Outlet	Outlet	Outlet	Outlet, Year	Outlet × Year
Number of clusters	18	18	18	18, 8	141
Observations	1,410	1,410	1,410	1,410	1,410

SEs in parentheses, clustered by newspaper in (1)–(3), two-way by newspaper and year in (4) (8 year clusters), by newspaper × year in (5). Randomization inference enumerates all 48,620 assignments of 9 left-wing labels across the 18 outlets. Wild cluster bootstrap-t at the newspaper level, Rademacher weights, null imposed, 1,999 replications. Stars follow the cluster-robust p, in (3) the randomization inference p. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$.

Table D3: **Robustness to Time-Varying Audience and Geographic Controls**

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Prob(Adv>0)						
Left-Wing Government × Left-Wing Outlet	0.183**	0.227**	0.188**	0.225**	0.174*	0.241**
	(0.082)	(0.098)	(0.082)	(0.088)	(0.084)	(0.092)
Panel B. Level (2022 USD)						
Left-Wing Government × Left-Wing Outlet	105,362*	112,534**	108,690*	118,576*	80,939**	97,738*
	(53,749)	(52,633)	(54,715)	(61,396)	(36,009)	(46,662)
Pre-2018 outlet characteristics × year FE						
Circulation share in above-median left-swing states		✓				✓
Circulation-weighted change in left vote share			✓			
Reader share outside AB/C+				✓		✓
Reader share under 35					✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Outlet × Ministry FE	✓	✓	✓	✓	✓	✓
Observations	1,410	1,410	1,410	1,410	1,410	1,410

Outlet characteristics are 2014–2017 averages. Left swing is the change in the left-wing presidential vote share, 2012–2018. SEs clustered at the newspaper level (18 clusters). $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$.

E Robustness for the Benchmarks

E.1 Prices, Placement, and the Source of the Reach Gap

Ministries pay more per unit of space than private advertisers do, and this appendix asks what that gap is and whether it drives the reach shortfall in Section 5. Measured against the price private buyers pay for the same space, a ministry pays 1.56 times as much at the median and 1.66 times as much in aggregate. These ratios compare each purchase with the price private buyers pay for the same space, and they are lower than the ratios of mean prices in Table 1, which compare the average price the government pays with the average list price. Two explanations compete. Ministries may purchase better placements, since a front page in color costs more than an inside page in black and white and the market price averages across both. Or they may pay more for the same product.

Placement cannot account for a gap of that size. Outlets quote their own market advertising prices by position and by color, and Table E1 reports what each combination costs relative to the median quote. The most expensive product available, an odd page in color, costs 1.13 times the median, and the cheapest, an even page in black and white,

costs 0.57 times. A ministry that purchased only the single most expensive placement on every card would therefore pay 13 percent above the middle of the market, against the 56 percent it actually pays. The purchase of premium placements explains at most a quarter of the gap.

Table E1: **Market Advertising Price per Square Centimeter by Position and Color**

	Quotes	Price per cm^2 (median)	Relative to the median quote
No fixed position, colour	213	116.2	1.13×
Odd page, colour	90	114.3	1.12×
Unspecified, colour	1167	108.9	1.06×
Even page, colour	180	94.2	0.92×
No fixed position, black and white	112	84.6	0.83×
Unspecified, black and white	551	75.9	0.74×
Odd page, black and white	40	64.4	0.63×
Even page, black and white	92	58.2	0.57×

Quotes drawn from the outlets' own published tariffs, collected from commercial sites and archived copies, restricted to those reporting a positive price per square centimeter. The final column divides each combination's median by the median across all quotes. Position labels follow the cards: odd and even refer to the page on which an advertisement runs, and no fixed position means the outlet places it at its discretion.

The remaining gap is concentrated where the recorded number of advertisements is least trustworthy. A cell recording one advertisement carries a median price ratio of 4.08 and a ninetieth percentile of 27.2, while a cell recording more than twenty carries a median of 0.96 and a ninetieth percentile of 1.72. Ministries that purchase at scale therefore pay almost exactly the private market rate. Single-advertisement cells hold 6.6 percent of the advertisements but 34 percent of the spending, which is what a systematic undercount of insertions would produce and what a genuine price premium would not, since no plausible placement costs twenty-seven times its market rate.

That distinction determines how much of the reach gap overpayment can explain, and Table E2 shows it explains almost none. Across all cells, pricing a ministry's own outlet choices at market rates would recover 96,588 readers per campaign. Restricting to cells recording at least three advertisements, where the price ratio approaches one, that figure falls to 4,422, and restricting to six or more it turns slightly negative. The misplacement

term moves in the opposite direction and stays close to half a million readers throughout.

Table E2: **Reach Gap Decomposition by Number of Advertisements Recorded**

	Delivered (readers)	Overpayment (readers)	Misplacement (readers)
All cells	178,486	109,817	439,227
Cells with 3 or more	136,565	11,607	469,239
Cells with 6 or more	101,530	-1,839	476,531

Readers per campaign at $\phi = 0.10$, averaged across campaigns. Delivered reach uses the recorded number of advertisements. The overpayment column prices a ministry's own outlet choices at the private market rate and reports the additional readers that would follow. The misplacement column reallocates the same spending to the outlets that reach the named audience most cheaply. Rows restrict the sample to cells recording at least the stated number of advertisements, retaining all cells with none.

Two further questions remain about prices, and they are separate from the level of the gap. The first is whether the gap is a gap for the *same* product, since the market benchmark averages across sizes and placements. The second is whether ideological alignment predicts what a ministry pays. Table E3 answers both by comparing two outlets carrying the same campaign at the same advertisement size, so the identity of the product is held fixed rather than averaged over.

Alignment does not move the price. The point estimates are small relative to their standard errors and mixed in sign, and this holds for the price paid, for the price relative to the outlet's published list price, and for the price per reader. With eighteen clusters the normal approximation is not reliable, so the p-values come from a wild cluster bootstrap with the null imposed, and they are larger still. The estimates are also conditional in a way the extensive-margin result is not. A price is observed only where a contract was placed, and alignment raises the probability of placement from 32.7 to 43.9 percent, so the sample the price is measured on is selected on a variable the treatment moves. Trimming the aligned distribution by that excess share gives Lee (2009) bounds, reported in the final row, and the identified interval contains zero. Favoritism operates on whom the government buys from and how much it spends, not on the unit price it pays.

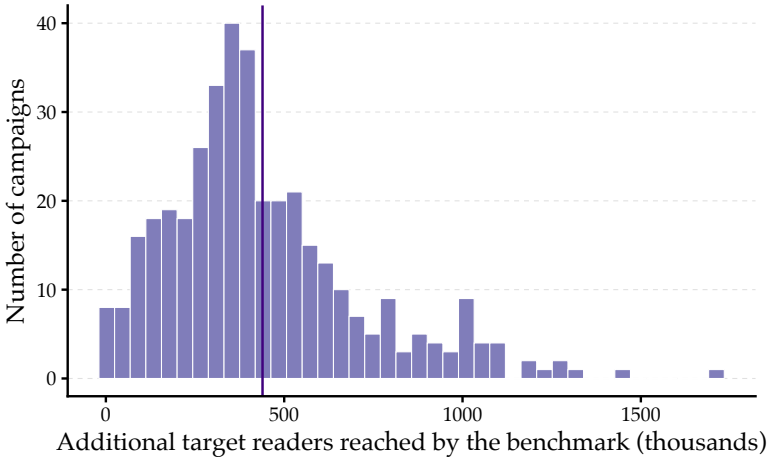
Table E3: **Price Paid to Aligned Outlets, Conditional on a Contract**

	$\log(p_{gov})$		$\log(p_{gov}/p_{mkt})$		$\log(p_{gov}/\text{circ})$	
	(1)	(2)	(3)	(4)	(5)	(6)
Left-Wing Government × Left-Wing Outlet	0.136	0.219	-0.038	0.022	-0.035	0.128
	(0.216)	(0.169)	(0.221)	(0.219)	(0.244)	(0.202)
	[0.544]	[0.358]	[0.835]	[0.929]	[0.978]	[0.728]
Outlet × Size FE	Yes	Yes	Yes	Yes	Yes	Yes
Campaign FE		Yes		Yes		Yes
Observations	8,597	8,597	8,597	8,597	8,597	8,597
Lee (2009) bounds on column (4): [-0.270, 0.278], trimming 25.4% of the aligned distribution.						

Note: *p<0.1; **p<0.05; ***p<0.01. Standard errors clustered on outlet in parentheses; wild cluster bootstrap p-values with the null imposed in brackets (2,000 Rademacher draws), which is the relevant inference with 18 clusters. The market price is the outlet-year published list price from the mediavyasa rate cards. A price is observed only where a contract was placed, and alignment raises the probability of placement, so the coefficient is not point identified; the Lee bounds give the sharp identified interval under the assumption that alignment never reduces the chance of placement.

The misplacement term is the one that does not depend on the insertion count. Both quantities it compares, what a ministry's own choices deliver at market prices and what the benchmark delivers, convert spending into insertions by dividing by the same price, so an undercount cancels. Ministries reach 275,073 readers with the outlets they chose and would reach 764,514 with the outlets that serve the campaign's audience, a factor of 2.8. That comparison stands whatever the recorded advertisement counts turn out to be.

Figure E1: **Difference in Unique Target Readers Reached, Benchmark Minus Observed**



One observation per campaign. Reach counts readers in the campaign's named groups who see the advertisement at least once, computed at $\phi = 0.10$. Observed reach converts spending into insertions using the advertisement counts ministries recorded, and benchmark reach converts it at the private market price. The dashed line marks zero and the solid line marks the mean difference.

E.2 Estimating Outlet-Level Copartisan Shares

I measure each outlet's copartisan reach using geographic variation in circulation. For each outlet j, year t, and state s, I compute the weight w_{js} as outlet j's share of its own total circulation that falls in state s:

$$w_{jst} = \frac{\text{circ}_{jst}}{\sum_{s'} \text{circ}_{js't}} \tag{6}$$

The copartisan share for party p in election year τ is then the circulation-weighted average of state-level vote shares:

$$\widehat{\text{Copartisan}}^{p}_{j\tau} = \sum_{s} w_{jst} \cdot v^{p}_{s\tau} \tag{7}$$

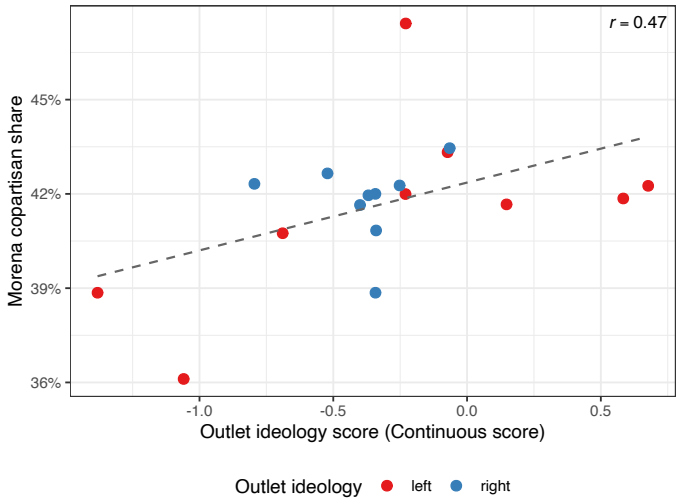
where $v^{p}_{s\tau}$ is the official vote share of party p in state s at election τ. Circulation data come from a balanced panel of regional edition figures for 18 national newspapers.[13]

A key assumption underlying the voter-targeting benchmark is that Morena and PRI

[13] Circulation is measured as copies per 10,000 inhabitants, obtained from the National Print Media Registry (PNMI). State-level vote shares come from official results published by the National Electoral Institute (INE) for the 2015, 2018, and 2021 midterm and presidential elections. Because advertising campaigns span multiple calendar years, I assign each year to the nearest election cycle: 2014–2016 to the 2015 midterm, 2017–2019 to the 2018 presidential election, and 2020–2022 to the 2021 midterm.

governments could, in principle, select outlets whose readership skews toward their own supporters. Figure E2 tests this by plotting each outlet's ideology score, a continuous measure of the partisan valence of its editorial language (Gentzkow et al., 2019), against its estimated Morena copartisan share in 2018. The ideological gap in copartisan reach is negligible. Left-wing and right-wing outlets attract nearly identical partisan audiences, and all 18 outlets fall within a 10-percentage-point band. The average Morena copartisan share is 41.6% among left-wing outlets and 41.8% among right-wing outlets. Voters in Mexico do not sort into newspapers by ideology, which limits the scope for outlet-level partisan targeting.

Figure E2: **Outlet Ideology Score and Morena Copartisan Share (2018)**

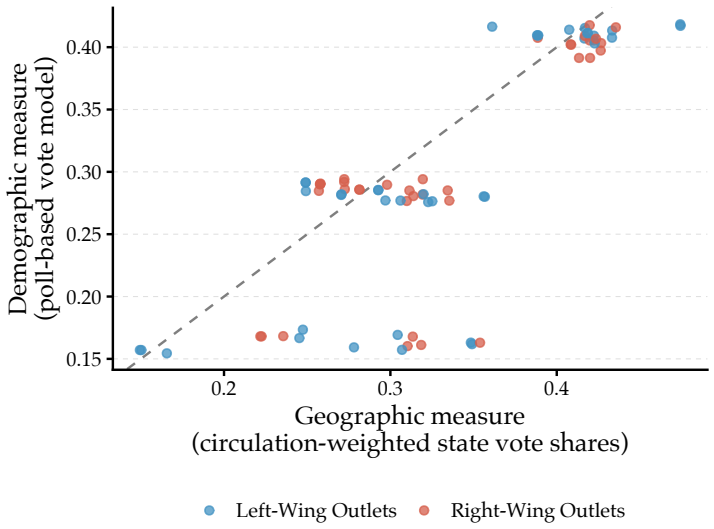


Each point is one of 18 national newspapers. The x-axis is the outlet's continuous ideology score from Gentzkow et al. (2019); more negative values indicate more left-leaning editorial language. The y-axis is the estimated share of the outlet's readers who supported Morena, computed as a circulation-weighted average of state-level Morena vote shares (2018 presidential election). The dashed line is an OLS fit. r denotes the Pearson correlation.

The geographic measure can be checked against an independent construction. Individual polling microdata registered with the National Electoral Institute records vote intention alongside state, gender, education, age and income; I estimate vote choice on those covariates and apply the fitted model to each outlet's registry audience profile, giving a demographic estimate of copartisan reach that shares no inputs with the geographic

measure beyond state of residence. Figure E3 plots the two against each other across the 76 outlet–years where both are available. They correlate at 0.80. On the demographic measure left-wing outlets reach a Morena share of 30.1 percent against 30.5 percent for right-wing outlets, a gap of −0.4 percentage points. On the geographic measure the figures are 33.3 and 33.4 percent, a gap of −0.2 points. The absence of an ideological gap in copartisan reach is therefore not an artifact of building the measure from circulation geography.

Figure E3: **Demographic and Geographic Estimates of Copartisan Reach**



Estimated Morena share of each outlet's readership, 76 outlet–years. The horizontal axis weights state-level official vote shares by the outlet's circulation across states. The vertical axis applies a vote-choice model estimated on polling microdata to the outlet's registry audience profile. The dashed line is the 45-degree line; colors denote the outlet's ideology classification from Section 3.2.

E.3 The Exposure Rate ϕ

The technology of Section 5 has one free parameter, the probability ϕ that a reader sees any one insertion. Rather than defend a value, I report the estimate across the range over which the budget constraint binds. As $\phi \to 0$ reach becomes linear in spending and the optimum is a corner placing the entire budget in the cheapest outlet per target reader, the maximum-concentration allocation. The limit $\phi \to 1$ is degenerate and is not reported.

As $\phi \to 1$ a single insertion saturates an outlet, so additional spending buys no further reach and the budget ceases to bind. The corresponding allocation spends only 57 percent of the average campaign budget and underspends in 251 of 383 campaigns, so a dollar deviation measured against it would mechanically contain the unspent remainder rather than a reallocation.

Table E4: **Intensive-Margin Deviation Across the Full Range of ϕ**

	$\phi \to 0$ (corner)	$\phi = 0.05$	$\phi = 0.10$	$\phi = 0.15$	$\phi = 0.90$
	(1)	(2)	(3)	(4)	(5)
Aligned (binary)	38,840	21,763**	20,876**	20,504**	19,072**
	(26,210)	(8,803)	(8,912)	(8,909)	(8,417)
Aligned (continuous)	38,440**	28,987***	28,749***	28,304***	26,299***
	(16,124)	(5,560)	(6,008)	(6,141)	(6,106)
Ad Campaign FE	✓	✓	✓	✓	✓
Outlet FE	✓	✓	✓	✓	✓
Observations	6,711	6,711	6,711	6,711	6,711

*Deviations from the cost-efficiency benchmark, in 2022 USD. SE clustered by outlet. ***$p<0.01$, **$p<0.05$, *$p<0.10$. Column (1) is the $\phi \to 0$ corner and columns (2) to (4) are interior. Column (5) sets $\phi = 0.90$, at which the benchmark leaves 4.7 percent of the average campaign budget unspent and underspends in 43 of the 383 campaigns. Column (3) reproduces Panel B, column (1) of Table 4.*

Table E5 reports the distribution across campaigns of the budget multiplier λ recovered in Section 5, of the reservation cost per target reader it implies, and of the market advertising prices those are compared against. The reservation cost sits inside the observed price distribution at every quantile, which is what produces interior solutions rather than corner ones.

Table E5: **Recovered Shadow Price and Reservation Cost Across Campaigns**

	p10	p25	Median	p75	p90
Shadow price λ (target readers per USD)	0.15	0.52	1.07	1.93	3.95
Reservation cost $\bar{c}$ (USD per 1,000 target readers)	27	55	99	203	719
Rate-card cost per 1,000 target readers	16	56	88	134	231

One observation per campaign, at $\phi = 0.10$. λ is recovered by bisection on the budget constraint under equation 3. The reservation cost is $\bar{c} = \kappa/\lambda$ with $\kappa = -\ln(1-\phi)$. The last row is the registry market advertising cost per thousand readers in the campaign's stated demographic, pooled across outlet–campaign cells.

The intensive-margin deviation is positive at every point in the bracket, as Table E4 reports, from $38,840 at the concentration limit to $20,504 at $\phi = 0.15$, and the continuous-score specification is significant at the five percent level at the corner and at the one percent level throughout the interior. The calibration of ϕ therefore does no work in the conclusion. Two quantities are more sensitive and should be read with that in mind. The number of outlets the efficient allocation uses rises from one at the concentration limit to 10.3 at $\phi = 0.10$ and 15.4 at the coverage limit, so the statement that campaigns are more concentrated than efficiency requires is qualitative rather than a point estimate. And the foregone-audience gap ranges from roughly three to five times delivered reach across the interior of the bracket.

E.4 Reader Sorting and the Voter-Targeting Benchmark

The voter-targeting benchmark in Section 5 builds copartisan reach from where an outlet circulates, so it misses any tendency of incumbent supporters to choose newspapers that share their politics. This appendix asks how strong that sorting would have to be before the result disappears. Writing ρ_{it} for the measured copartisan share, I tilt it by

$$\text{logit}(\tilde{\rho}_{it}) \;=\; \text{logit}(\rho_{it}) + \tau\, z_i, \qquad z_i = \tfrac{1}{2} \text{ if aligned, } -\tfrac{1}{2} \text{ otherwise} \tag{8}$$

so that e^{τ} scales the odds that an incumbent supporter reads an aligned rather than a misaligned newspaper. I apply the tilt in log-odds so that it means the same thing at any base rate, which matters because the incumbent's vote share differs sharply across the two administrations. I then rebuild the voter-targeting benchmark with $\tilde{\rho}_{it}$ and estimate equation 4 again at each value.

How much sorting the result tolerates depends on the scale I measure the deviation on. In Table E6 I raise the assumed sorting from none in column (1) to a multiplier of 4.5 in column (6), rebuilding the voter-targeting benchmark and re-estimating equation 4 at

Table E6: **Deviations from the Voter-Targeting Benchmark Under Assumed Reader Sorting**

	Assumed sorting, odds ratio e^{τ}					
	1.00	1.28	1.65	2.12	2.72	4.48
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Deviation in 2022 dollars						
Aligned	22,198**	16,035	9,964	4,136	−1,274	−10,686
	(9,865)	(9,567)	(9,285)	(9,032)	(8,835)	(8,609)
Ideology score	31,038***	27,015***	23,003***	19,127**	15,543*	9,408
	(6,863)	(7,045)	(7,349)	(7,731)	(8,175)	(9,120)
Panel B. Deviation as share of campaign budget (pp)						
Aligned	3.01*	1.64	0.26	−1.08	−2.35	−4.65**
	(1.71)	(1.63)	(1.59)	(1.59)	(1.65)	(1.90)
Ideology score	4.08**	3.14*	2.18	1.25	0.39	−1.14
	(1.65)	(1.75)	(1.92)	(2.15)	(2.41)	(2.98)
Implied aligned−misaligned gap (pp)	−0.9	+3.2	+7.4	+11.5	+15.6	+23.8
Ad Campaign FE	✓	✓	✓	✓	✓	✓
Outlet FE	✓	✓	✓	✓	✓	✓
Observations	6,711	6,711	6,711	6,711	6,711	6,711

Intensive-margin estimates of equation 4, with the copartisan share replaced by $\tilde{\rho}_{it}$ from equation 8. Panel A takes the deviation in 2022 dollars and Panel B divides it by the campaign's total budget. Column (1) uses the measured share. The measured baseline odds ratio does not equal one, so the headings report the assumed multiplier applied to it rather than the resulting ratio. The final row reports the implied difference in mean copartisan share between aligned and misaligned outlets, which depends jointly on the multiplier and on the base rate. Standard errors clustered by outlet. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.10$.

each step. Panel A measures the deviation in dollars and survives the most sorting, since the continuous estimate stays significant at the five percent level out to a multiplier of 2.1 and at ten percent out to 2.7. Panel B divides each deviation by its campaign budget and survives less, since there the continuous estimate holds at ten percent only to 1.3 and the binary estimate loses significance as soon as I impose any sorting at all. The binary estimate turns negative in both panels once the multiplier passes 2.1, and by 4.5 the sign reverses at the one percent level in Panel B, so sorting that strong would make the benchmark want to buy aligned newspapers more heavily than ministries did. Taking the share scale as the binding one, the result survives as long as the odds of an incumbent supporter reading an aligned rather than a misaligned newspaper stay within roughly 1.3 times what I measure, which would put 24 percent of an aligned outlet's readers on the incumbent's side against 21 percent of a misaligned outlet's.

E.5 Reader Demographics and the Allocation

The benchmark treats every reader in a campaign's named groups as equally valuable. Gentzkow et al. (2024) instead allow each viewer an intrinsic value to advertisers, which raises the possibility that the government weights demographic groups differently and that the apparent misallocation reflects those weights rather than ideology. This appendix asks what weights would be required.

Replace the audience with $a_i(w) = N_i \sum_g w_g r_{ig}$, where r_{ig} is outlet i's share of readers in group g across three income and six age categories, and choose $w \geq 0$ to minimize the squared distance between the resulting optimal allocation and the observed one.

The weights are not identified, for a reason that is exact rather than numerical. Income shares sum to one and age shares sum to one for every outlet, so adding a constant to all three income weights and subtracting the same constant from all six age weights leaves $a_i(w)$ unchanged for every outlet, verified to machine precision at any magnitude. The objective is correspondingly flat. Fixing any single group's weight at zero and re-

optimizing the rest costs at most 2.7 percent of the objective, and seven of the nine cost under half a percent. A further obstacle is that r_{ig} is time-invariant and outlet-specific, so $a_i(w)$ does not vary across campaigns and cannot in principle reproduce campaign-specific targeting. The estimated weights are therefore not reported.

What the exercise does deliver is a comparison. Table E7 reports how well each specification reproduces the observed allocation. The informative column is the last, the correlation between predicted and observed *within-campaign* shares, since the correlation in levels is largely driven by campaign budget. Splitting each budget equally across outlets, with no model at all, attains a level correlation of 0.49. Every audience-based specification predicts within-campaign allocation negatively. A single parameter loading circulation on outlet ideology reverses that, and outperforms nine free demographic weights.

Table E7: **Fit of Alternative Audience Weightings to the Observed Allocation**

	Objective ($\times 10^{13}$) (1)	**Correlation, levels** (2)	**Correlation, within-campaign shares** (3)
Equal weights, $a_i \propto N_i$	9.604	0.453	−0.103
Benchmark audience weight	9.627	0.454	−0.083
Best demographic weights (9 parameters)	9.273	0.479	−0.056
Circulation × ideology (1 parameter)	9.065	0.505	+0.110
Demographic weights + ideology	8.954	0.508	+0.062

Each row solves equation 2 campaign by campaign with the stated audience weight and compares the result to observed spending. The objective is the summed squared deviation across outlet–campaign cells. The ideology specification sets $a_i = N_i \exp(\theta \cdot \text{score}_i)$ *with* θ *estimated freely; the fitted value is* $\theta = 1.28$. *Column (3) computes the correlation of predicted and observed spending shares within each campaign, then averages.*

E.6 Robustness to Low-Cost Titles

The outlets the cost-efficiency benchmark most wants to buy are those with the lowest registry market advertising cost per thousand readers in the campaign's stated demographic, and the three cheapest by that measure are high-circulation popular and sports

titles, namely La Prensa at $10.7, Esto at $16.0 and Ovaciones at $22.9, against $53.9 for El Universal and $61.7 for El Sol de México. Two concerns follow. Published list prices for these titles may understate the prices at which the government could actually transact, and registry audience shares may overstate how far a campaign placed in a sports paper reaches the demographic it names. Either would inflate the measured gap between the observed allocation and the benchmark.

Table E8 re-estimates equation 4 dropping these outlets. Column (2) removes the two sports titles and column (3) removes all three cheapest. Dropping an outlet removes it from the choice set, so both benchmarks are re-solved on the retained outlets and each campaign's budget is reallocated only among them. Every estimate rises rather than falls. On the cost-efficiency benchmark the binary estimate goes from $20,876 in the full sample to $24,286 and $27,183, and the continuous estimate from $28,749 to $29,364 and $30,649; the voter-targeting estimates move in the same direction, from $22,198 to $26,239 and $30,181 on the binary measure. Standard errors on the binary specifications widen as outlets are dropped, which is expected with eighteen clusters, but the continuous specifications remain significant at the one percent level throughout. The result is not produced by the benchmark's preference for cheap high-circulation titles.

Table E8: **Deviations from the Benchmarks Excluding Low-Cost Titles**

	Full sample (1)	Excl. sports titles (2)	Excl. three cheapest (3)
Cost-efficiency, binary alignment	20,876**	24,286**	27,183**
	(8,912)	(10,463)	(11,606)
Cost-efficiency, continuous score	28,749***	29,364***	30,649***
	(6,008)	(5,918)	(5,536)
Voter-targeting, binary alignment	22,198**	26,239**	30,181**
	(9,865)	(12,021)	(13,483)
Voter-targeting, continuous score	31,038***	32,249***	34,231***
	(6,863)	(6,829)	(6,335)
Ad Campaign FE	✓	✓	✓
Outlet FE	✓	✓	✓
Observations	6,711	5,945	5,562

Intensive-margin estimates of equation 4 in 2022 USD, at $\phi = 0.10$. Column (2) excludes Esto and Ovaciones; column (3) additionally excludes La Prensa. Both benchmarks are re-solved campaign by campaign on the retained outlets. SE clustered by outlet. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.10$.

E.7 Audience Controls and the Benchmark as a Control

Table E9 adds pre-2018 outlet characteristics interacted with year fixed effects to equation 4. The coefficient on alignment ranges from 0.106 to 0.139 on the extensive margin and from \$13,932 to \$20,876 on the intensive margin. Table E10 takes the observed allocation as the outcome and controls for the benchmark allocation directly. The coefficient on alignment ranges from 0.084 to 0.088 on the extensive margin and from \$19,652 to \$20,435 on the intensive margin.

Table E9: **Benchmark Deviations with Audience and Geographic Controls**

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. $\mathbf{1}[s^{obs}_{ijt} > 0] - \mathbf{1}[s^{*}_{ijt} > 0]$						
Alignment	0.139***	0.123***	0.134***	0.122**	0.132***	0.106**
	(0.039)	(0.041)	(0.039)	(0.043)	(0.040)	(0.040)
Panel B. $s^{obs}_{ijt} - s^{*}_{ijt}$ *(2022 USD)*						
Alignment	20,876**	19,202**	20,678**	19,656*	16,242**	13,932*
	(8,912)	(7,247)	(8,805)	(9,669)	(6,637)	(7,068)
Pre-2018 outlet characteristics × year FE						
Circulation share in above-median left-swing states		✓				✓
Circulation-weighted change in left vote share			✓			
Reader share outside AB/C+				✓		✓
Reader share under 35					✓	✓
Ad Campaign FE	✓	✓	✓	✓	✓	✓
Outlet FE	✓	✓	✓	✓	✓	✓
Observations	6,711	6,711	6,711	6,711	6,711	6,711

Outlet characteristics are 2014–2017 averages. Left swing is the change in the left-wing presidential vote share, 2012–2018. SEs clustered at the newspaper level (18 clusters). $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$.

Table E10: **The Benchmark as a Control**

	Observed Allocation			**Deviation**
	(1)	(2)	(3)	(4)
Panel A. Extensive margin				
Alignment	0.084**	0.088***	0.087***	0.139***
	(0.030)	(0.029)	(0.029)	(0.039)
$\mathbf{1}[s^{*}_{ijt} > 0]$		0.073**		
		(0.033)		
Panel B. Intensive margin (2022 USD)				
Alignment	19,652**	20,435**	20,305**	20,876**
	(8,363)	(8,710)	(8,413)	(8,912)
s^{*}_{ijt}		0.639**	0.751**	
		(0.229)	(0.295)	
Benchmark bins ($s^{*} = 0$, deciles of $s^{*} > 0$)			✓	
Ad Campaign FE	✓	✓	✓	✓
Outlet FE	✓	✓	✓	✓
Observations	6,711	6,711	6,711	6,711

Columns (1)–(3) take the observed allocation as the outcome. Column (4) takes the deviation from the benchmark as the outcome. SEs clustered at the newspaper level (18 clusters). $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$.

F Robustness for the Downstream Effects

F.1 Topic Model and the Hard News Classification

The classification of articles into hard and soft news in Section 6 comes from a structural topic model (Roberts et al., 2014) estimated on a stratified sample of up to 1,000 articles per outlet and year, or 162,000 articles in total. The sample excludes advertising, paid inserts, obituaries, cartoons, legal notices, tables, open letters, and articles shorter than 100 characters. Preprocessing joins the headline to the body and removes accents, punctuation, numbers, short tokens, standard Spanish stopwords, and 56 terms common to Mexican newsrooms, such as attribution verbs and wire bylines. The remaining tokens are stemmed, and stems that appear in fewer than 10 documents are dropped (Roberts et al., 2019).

The model has 40 topics. Topic prevalence depends on outlet indicators, author type, and a spline in time, and topic content depends on outlet ideology. I chose the number of topics on a pilot sample by comparing models with 10 to 50 topics on held-out likelihood, semantic coherence, exclusivity, and residual dispersion. Held-out likelihood, exclusivity, and residual dispersion improve with the number of topics and level off after 30, while semantic coherence declines, so I set the number of topics to 40.

I apply the estimated topic-word distributions to all 4,595,189 articles published between 2014 and 2022. Each article is assigned the topic with the largest proportion, and its classification as hard or soft news follows from that topic. The counts of hard and soft news in Table 5 exclude wire copy. Hard news accounts for 68.2 percent of articles.

To label the topics, I provided a large language model (claude-opus-4-6) with the ten words of highest probability and the ten words of highest FREX score for each topic, the words most distinctive to left-wing and to right-wing outlets, and the mean proportion of each topic, together with definitions of hard and soft news that follow Cagé (2020) and Angelucci and Cagé (2019), with crime classified as hard news. I reviewed every label

against its word list. The review added one topic omitted from the model output, which describes personal and family life and is classified as soft news. The final classification contains 26 hard topics and 14 soft topics, listed in Table F1. Six topics capture writing style rather than subject, such as official announcements and opinion writing, and excluding them raises the share of hard news from 68.2 to 69.8 percent.

I validate the classification against the newspaper section in which each article appeared, which is recorded independently of the model. I map the 89 most common section names to hard or soft news and exclude sections, such as the front page or opinion, that do not identify a subject. In 2018 and 2019, the mapping covers 79.7 percent of articles. The two classifications agree for 87.0 percent of these articles, for 88.0 percent after excluding the six style topics, and for 91.5 percent of articles whose dominant topic accounts for at least half of the article.

Table F1: **The Forty Estimated Topics and Their Hard News Assignment**

Topic	Label	Category	Type	Share (%)
1	Aviation & Auto Industry	Infrastructure	Hard	1.5
2	Development & Innovation Policy†	Development	Hard	2.9
3	Personal & Family Life	Human Interest	Soft	2.0
4	Catholic Church & Religion	Culture	Soft	0.9
5	Presidential Politics	National Politics	Hard	2.2
6	Events & Announcements†	Culture	Soft	0.9
7	Public Finance & Taxation	Fiscal	Hard	3.1
8	Venezuela & Cuba Politics	International	Hard	1.0
9	Soccer Match Reporting	Sports	Soft	1.7
10	Brazil Politics	International	Hard	0.3
11	Left-Right Ideology & Populism	International	Hard	2.3
12	US & International Sports	Sports	Soft	3.0
13	Education & Teachers Unions	Education	Hard	2.0
14	Human Rights & Military	National Politics	Hard	2.2
15	Opinion & Informal Commentary†	Opinion	Soft	1.4
16	Macroeconomics & Financial Markets	Economy	Hard	4.4
17	Drug Cartel Leaders & Crime	Crime & Security	Hard	0.9
18	Technology & Science Commentary†	Opinion	Soft	1.3
19	Congress & Legislation	National Politics	Hard	3.4
20	European Soccer (Champions League)	Sports	Soft	2.6
21	State & Municipal Government	Local	Hard	1.0
22	Business & Tech Companies	Firms	Hard	2.1
23	Public Health & Food Policy	Health Policy	Hard	2.3
24	Arts & Literature	Culture	Soft	3.0
25	COVID-19 Pandemic	Health Policy	Hard	3.1
26	Political Opinion & Analysis†	Opinion	Soft	0.6
27	Courts & Judiciary	Justice	Hard	4.1
28	Electoral Politics	Elections	Hard	4.1
29	Crime & Public Security	Crime & Security	Hard	3.3
30	Mexico City Infrastructure	Local	Hard	4.9
31	Telecom & Digital Regulation	Telecom	Hard	2.4
32	Formula 1 & Olympic Sports	Sports	Soft	2.2
33	Energy & Trade Policy	Energy	Hard	2.5
34	Music & Film	Entertainment	Soft	6.1
35	International Conflict & Terrorism	International	Hard	4.0
36	Tourism & Gastronomy	Tourism	Soft	1.8
37	Statistics & Data Reporting†	Economy	Hard	1.5
38	Gender & Women's Rights	Gender	Hard	1.6
39	Liga MX Soccer	Sports	Soft	4.3
40	US Politics & Migration	International	Hard	4.9

Each row is one topic of the structural topic model with $K = 40$ estimated on the stratified training sample of the 18 national outlets, 2014 to 2022. Label and Category record the labeling step described above, and Type records the hard or soft assignment used throughout the paper. Share gives the percentage of the 4,595,189 scored articles for which the topic carries the largest proportion, and the column sums to 100. A dagger marks the six topics whose leading words describe writing register instead of subject. These six keep their assignment, and the robustness check that drops them raises the hard news share from 68.2 to 69.8 percent.

F.2 Constructing News Events and the Within-Event Measures

The quality measures in Section 6 compare outlets that report on the same story, which requires grouping articles into news events. The corpus contains the 3,945,226 news articles that the 18 outlets published between 2014 and 2022. Each article, including its headline, is lowercased, stripped of accents and non-letter characters, and represented as a term-frequency inverse-document-frequency vector over a vocabulary fixed across the sample. The similarity between two articles is the cosine of their vectors.

Following Cagé et al. (2020) and Durante et al. (2026), I form events within a three-day window that advances two days at a time. Within each window, single-linkage hierarchical clustering joins two articles when a chain of articles connects them at a cosine similarity of at least 0.32. Clusters in overlapping windows that share an article are merged, so an event can last longer than three days. The procedure assigns 2,159,555 articles, or 54.7 percent of the corpus, to 410,916 events (Table F2).

I chose the linkage rule and the similarity threshold using 38 dated events from the Wikipedia timelines for 2018 and 2019, selected before any clustering output was inspected. I compared 22 combinations of linkage rule and threshold against six criteria fixed in advance, which limit the fragmentation of each external event, the share of articles in the largest cluster, and the duration of events. Two specifications satisfy all six criteria, and I use the one with the smaller largest cluster. Under this specification, the main cluster of 34 of the 37 matched events, or 91.9 percent, spans at least two outlets, which is comparable to the 92 percent recovery rate that Cagé et al. (2020) report for Europe Media Monitor stories.

An event enters the estimation sample if at least three outlets published on it, which leaves 193,066 events and 1,690,436 articles. An outlet that did not cover an event enters with a value of zero. Events covered by one or two outlets are similar in composition to those retained, with a hard news share of 67.1 percent against 70.7 percent, and drop rates by subject range from 40.6 percent for national politics to 60.9 percent for culture.

A factual element is a person, organization, location, or other named entity identified by the Spanish news model of spaCy, or a number with at least two digits. I extract elements from the headline and the first 4,000 characters of each article, after excluding wire copy, and count each element once per outlet and event. The median event contains 81 distinct elements, and the median outlet contributes 28 of them.

Three outcomes enter the estimation. Coverage equals one when outlet j published at least one article on event e. Scoops equals one when the first own article of outlet j on event e appeared on the first day of the event. It is defined for outlets that covered the event with their own reporting, for events lasting more than one day (455,313 observations). Because the treatment affects coverage, the scoops estimate conditions on an outcome of the treatment. Unique entities is the number of distinct factual elements that outlet j contributed to event e, set to zero when the outlet contributed none. It is defined for the 153,109 events with at least one extracted element (2,700,888 observations).

Table F2: **The Event Panel and the Element Panel**

	Value
Panel A. News events, 2014–2022	
Articles in the corpus	3,945,226
Articles assigned to an event	2,159,555
Share of articles assigned to an event	54.7%
Events formed	410,916
Events covered by at least three outlets	193,066
Articles in those events	1,690,436
Articles per event, mean / median	8.76 / 5
Outlets per event, mean / median	5.49 / 4
Events lasting a single day	57.5%
Event duration, 90th / 99th percentile	2 / 7 days
Outlet–event observations	3,475,188
Share covered	30.5%
Panel B. Factual elements	
Wire articles excluded before extraction	329,575
Articles parsed	1,360,859
Entity and number mentions extracted	37,836,207
Articles carrying at least three elements	1,303,858
Events retained	153,109
Outlet–event observations with elements	823,169
Distinct elements per event, median	81
Distinct elements per outlet–event, mean / median	39.1 / 28
Panel C. External event recovery, 2018–2019	
External events tested	38
Events matched by headline	37
Dominant cluster spans at least two outlets	91.9%
Mean concentration in the largest cluster	82.3%
Median clusters per external event	2

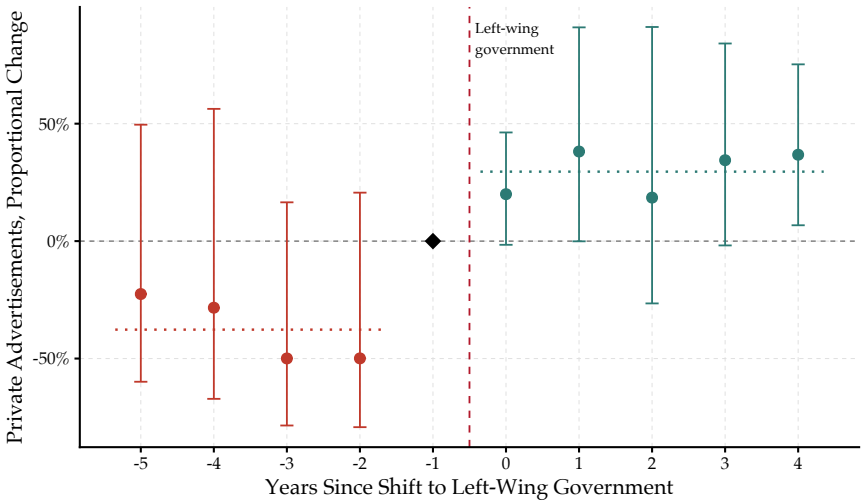
Panel A covers the 18 outlets from 1 January 2014 through 31 December 2022. Events come from single-linkage agglomerative clustering of term-frequency inverse-document-frequency vectors, cut at a cosine distance of 0.68 inside a three-day window that advances two days at a time. Panel B covers the same events after wire copy, articles carrying fewer than three elements, and events left with fewer than three covering outlets leave the sample. Elements are named entities from spaCy's `es_core_news_sm` pipeline together with numbers of at least two digits, counted once per outlet within an event. Panel C compares the event set against 38 events dated from the Wikipedia timelines for 2018 and 2019, matched on proper nouns in the headline.

F.3 Revenue, Newsroom, and Circulation

F.3.1 Revenue Shock to Outlets

Government advertising was not the only revenue that could have moved at the transition. Figure F1 runs the event study of Section 6 by Poisson PPML with the number of private advertisements an outlet carries as the outcome. Left-wing outlets carry more private advertising than right-wing outlets in every year after the transition, by 20 percent in the transition year and 38 percent in the year that follows, and the difference stays positive through 2022. The pre-transition coefficients lie below zero, and none of them separates from zero. The panel holds 18 outlets and 178 outlet–years. The pooled post-transition coefficient of 0.451, or 57 percent, carries a standard error of 0.290, and adding the number of days of coverage as an offset shrinks it to 0.247, or 28 percent, with a standard error of 0.215. The sign holds across both specifications and neither estimate separates from zero.

Figure F1: **Event Study of Private Advertising by Outlet Ideology**



Proportional change $\exp(\hat{\beta}) - 1$ on the interaction of a left-wing outlet indicator with each year relative to the 2018 transition, from a Poisson PPML of an outlet's private advertisement count on outlet and year fixed effects. The year before the transition is the omitted category, marked by the diamond. Bars are 95 percent confidence intervals from standard errors clustered by outlet across 18 clusters. Dotted lines mark the average coefficient before and after the transition. Outlet ideology follows the classification of Section 3.2.

F.3.2 Newsroom Size

Table F3 decomposes the articles outlets write themselves into finer content categories, estimating the Poisson specification of Table 5 on the monthly count of non-wire articles within each category.

Table F3: **Effect of Left-Wing Government on News Production by Content Type**

	Left-Wing Gov × Left-Wing Outlet	SE	$\exp(\hat{\beta}) - 1$	Mean pre-2018 (Left)	Mean pre-2018 (Right)
National Politics	0.334**	(0.122)	40%	122.9	186.3
Elections	0.334***	(0.092)	40%	39.1	60.9
Crime and Security	0.510***	(0.161)	67%	95.1	161.4
Justice	0.393***	(0.083)	48%	48.5	74.7
Economy	0.419***	(0.117)	52%	68.3	161.3
International	0.489***	(0.069)	63%	117.4	188.5
Health Policy	0.480***	(0.165)	62%	37.5	70.9
Local	0.447**	(0.164)	56%	88.2	174.8
Sports	0.379	(0.219)	46%	180.6	230.9
Culture	0.376**	(0.138)	46%	116.0	204.4
Entertainment	0.173	(0.185)	19%	39.8	80.8

Note: *p<0.1; **p<0.05; ***p<0.01. SEs clustered at the outlet level, with p-values from a t distribution with 17 degrees of freedom. Outlet–month panel, 2014–2022, N = 1,908. Each row is a separate Poisson PPML of the monthly count of non-wire articles in that category with outlet and year-month fixed effects. Means cover the period before the transition and are reported in levels.

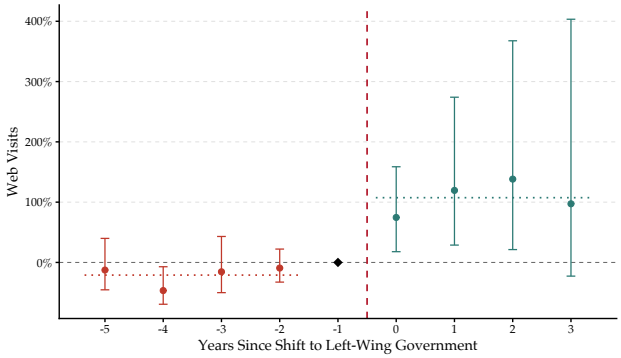
Left-wing outlets increase news production in every category relative to right-wing outlets. The gains are largest in crime and security (67 percent), international news (63 percent), and health policy (62 percent), followed by local news (56 percent), the economy (52 percent), and justice (48 percent). National politics and elections each rise by 40 percent. Among soft news categories, culture rises by 46 percent, the 46 percent estimate for sports separates from zero only at the 10.1 percent level, and the 19 percent estimate for entertainment does not separate from zero.

F.3.3 Web Visits

Print circulation is reported by the outlets themselves. Figure F2 instead uses monthly organic visits to each outlet's website from SEMrush. Web visits rise 138 percent at left-

wing outlets relative to right-wing outlets after the transition, with a Poisson coefficient of 0.868, a standard error of 0.245, and a p-value of 0.003.

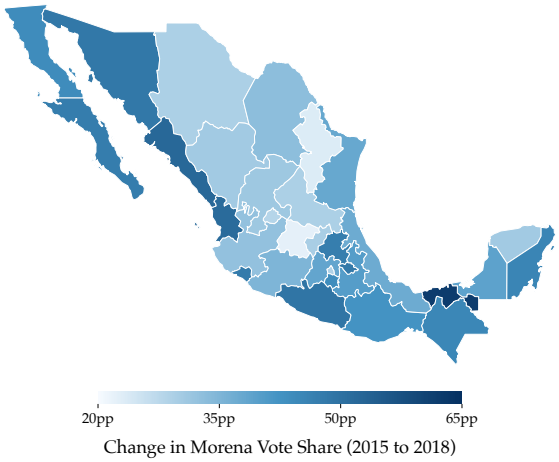
Figure F2: **Web Visits by Outlet Ideology**



Source: SEMrush monthly organic visits, 2014–2022. Poisson PPML with outlet and month fixed effects. The figure plots $\exp(\hat{\beta}) - 1$ with 95 percent confidence intervals. Standard errors clustered by outlet, with critical values from a t distribution with 17 degrees of freedom.

F.3.4 Circulation

Figure F3: **Change in Morena Vote Share by State (2015–2018)**



Change in Morena's vote share by state between the 2015 midterm and the 2018 presidential election, in percentage points. Darker shades indicate larger increases. States above the median change are classified as high-shift states ($D_s = 1$) in the triple difference-in-differences specification.